

Analytical and Machine Learning Framework for Predicting the Impact of LPBF Parameters on the Hardness of Inconel™ 718

Mohsen Dehghanpour Abyaneh

`mohsen.dehghanpour@polito.it`

Politecnico di Torino <https://orcid.org/0009-0001-7794-1509>

Raffaella Sesana

Politecnico di Torino

Mohammad Sadegh Javadi

Amirkabir University of Technology

Parviz Narimani

University of Tehran

Matteo Crachi

Politecnico di Torino

Antonio Caraviello

Sophia High Tech s.r.l.

Research Article

Keywords: Laser Powder Bed Fusion, Inconel™ 718, Hardness, Machine Learning, Formula evaluation

Posted Date: September 19th, 2025

DOI: <https://doi.org/10.21203/rs.3.rs-7648647/v1>

License:   This work is licensed under a Creative Commons Attribution 4.0 International License.

[Read Full License](#)

Additional Declarations: The authors declare no competing interests.

Analytical and Machine-Learning Framework for Predicting the Impact of LPBF Parameters on the Hardness of Inconel™ 718

Mohsen Dehghanpour Abyaneh ^{1,2*}, Raffaella Sesana ¹, Mohammad Sadegh Javadi ³, Parviz Narimani ⁴, Matteo Crachi ¹ and Antonio Caraviello ⁵

1. Department of Mechanical and Aerospace Engineering (DIMEAS), Politecnico Di Torino, 10129 Torino, Italy;
 2. TN ITALY, Global Manufacturing Organization, Corso Torino 378, 10064 Pinerolo, Torino, Italy.
 3. Department of Mechanical Engineering, Amirkabir University of Technology, 424, Hafez Ave., Tehran, P.O.B. 158754413, 15875-4413, Iran; javadims@aut.ac.ir
 4. School of Mechanical Engineering, College of Engineering, University of Tehran, Tehran, Iran; parviz.narimani@ut.ac.ir
 5. Sophia High Tech s.r.l., Via Malatesta 30 A, Napoli, 80049, Italy; antonio.caraviello@sophiahightech.com
- * Correspondence: mohsen.dehghanpour@polito.it

Abstract

In Laser Powder Bed Fusion, a layer-by-layer melting of metal powder takes place and is specifically suited to high-performance applications in advanced technologies using Inconel™ 718. ANOVA and hardness measurements identified relationships that cannot be predicted using conventional statistical methods. Particle Swarm Optimization and Genetic Algorithm were proposed to define the relationships between the input and output data. Although the mean accuracy was high (97% for Particle Swarm Optimization and 91% for Genetic Algorithm), the drawbacks were calculation variation and sensitivity to parameter changes. The prediction of hardness was then done using five regression models, such as Support Vector Machine, Gaussian Process Regression (GPR), Single-Layer and Deep-Layer Artificial Neural Network (ANN), and Random Tree (RT). The low R^2 values were observed in the initial implementation with a mean accuracy of 90%. The linear method of optimizing output data increased R^2 values up to and beyond 0.9 with high average accuracy. GPR and single-layer ANN performed best in terms of training results. A rollback process was implemented on the test results. GPR and ANN displayed the best results with the highest R^2 (0.99) and MAPE (1.3%) values on the testing data, which proved them as the best solutions.

Keywords

Laser Powder Bed Fusion . Inconel™ 718 . Hardness . Machine Learning . Formula evaluation

1 Introduction

The potential of metal additive manufacturing (AM), particularly Laser powder bed fusion (LPBF), to change manufacturing through the ability to produce complex geometries with tailored properties has now been realised to a large extent. Laser power, scanning speed as well as the hatching distance are the most commonly utilized parameters in optimization of processes. These

parameters have effects on volumetric energy density (ED), which determines the mechanical characteristics of the parts and rough surfaces (Ref 1, 2). ED and the scanning process determine the profile of temperature during the printing process, which determines the hardness of components. These interactions are optimizable using statistical tools and machine-learning (ML) algorithms, to have better mechanical processes.

The most commonly used nickel alloys when going through LPBF additive manufacturing are InconelTM 625 and InconelTM 718 to create aerospace parts. These materials have a high ability to resist corrosive media, intense temperatures, and creep, which is another benefit of them (Ref 3). Because InconelTM 718 can ensure the finest mechanical performance at high temperatures, it is frequently the ideal option for complex components such as liquid rocket engine components (Ref 4, 5, 6). A detailed overview of the use of InconelTM 718 in the production of aeronautical components is provided in (Ref 7). In the case of InconelTM 718, heating and cooling methods implemented in the deposition process of LPBF enhance the hardness of AM parts compared to the conventional methods (Ref 8).

The AM process has a large influence on the modeling of structural behaviors because the material mechanical properties determine the estimation of structural behaviors (Ref 9, 10). Hardness measurement is a simple and cost-effective way to estimate mechanical behavior. Hardness is referred to as an extrinsic material feature, which is a measure of the material's resistance to plastic deformation when penetrated using an indenter tool on the surface of components (Ref 6, 11). The hardness tests are either non-destructive or semi-destructive, and the hardness test results would prove to show a linear relationship with the mechanical parameters, such as the yield strength, tensile strength, and the Young modulus (Ref 12, 13). In (Ref 14), the paper discussed the experimental investigation of the influences of the deposition factors on the mechanical hardness and structural properties of InconelTM 718.

Any mechanical property, such as hardness, is hard to predict, technologically speaking, because of the complex interplay of process parameters (Ref 15). The ML regression models have been highly effective in the response to the challenge because of their ability to effectively use the data to apply a given estimate to a set of specific qualities in an accurate fashion and have been widely engaged to predict metal 3D printing mechanical properties (Ref 16). Such models facilitate optimization of systems to enhance the aspect of printing parts. Most of the predictions deal with only those properties that influence performance, which include hardness, tensile strength, and yield strength (Ref 17). A publication explores the role of ML in predicting mechanical properties in composite materials like yield strength, ultimate tensile strength, elastic modulus, elongation, hardness, and surface roughness in AM materials and components. It emphasizes the presence of the powerful effect of quality data and the combination of ML models and real-time monitoring systems as the crucial pressure of increasing predictability (Ref 8). Substantial ML studies have been performed on the topic of Mg alloys and have indicated that neural networks (NN), random forests (RF), and gradient-boosting (GB) trees are effective in the prediction of mechanical properties, including ultimate tensile strength, elongation, yield strength, and hardness. The other analysis in steel tubes involved the use of NNs, RF, and gradient-boosting trees to predict yield strength, ultimate tensile strength, and hardness. All of which were very precise and thus indicated the promise of ML in predicting the mechanical properties of process parameters (Ref 18, 19). In recent studies on Friction stir welding (FSW) joints of aluminum alloys, the prediction of mechanical characteristics of various temperature conditions using various ML models, like Support Vector Regression (SVR) and Gaussian Process Regression (GPR), has been examined. In this direction, the studies can be seen as the embodiment of the transformative contribution that

ML can make in terms of material science as a means of enhanced prediction and optimization of mechanical properties (Ref 20). In a review (Ref 21), ML methods of predicting mechanical properties of AM titanium alloys with a particular focus on the influence of processing factors are provided. This study highlights the fact that ML has the potential to transform materials science by making improved designs of materials and making more accurate predictions more straightforward, and shows that these techniques could be potentially applied to the optimization of process parameters accordingly. In addition, Xiong et al. (Ref 22) applied five ML algorithms to predict mechanical properties in steels. Among these, RF regression showed the best predictive performance, again underlining the critical role of feature selection in model accuracy.

In one of the articles on the use of ML, the role of processing parameters in the properties and characteristics of the AM components was considered. MechProNet data incorporates the data about AM processing conditions, machines, materials, and resulting mechanical properties (Ref 23, 24). To predict the mechanical properties of AM metals, the possibility to apply different ML methods and model the effects of process parameters was discussed in (Ref 23, 25). Such works also prove that, in this course, there are both challenges and opportunities, and they clarify that it is necessary to consider the quality of data and the interpretability of models. In (Ref 26, 27), publications revealed the potential of ML to predict the mechanical properties of additively manufactured parts such as the yield strength, ultimate tensile strength, and the hardness. Gu et al. (Ref 25) examined the prediction of mechanical parameters in LPBF through ML models where data quality and model interpretability were marked as significant. The aims of such solutions are to provide quality and reliability of the end-created elements due to merging ML and real-time control systems, and also to optimize the process of manufacturing. Luo et al. (Ref 28) investigated the prediction of mechanical properties of Ti-6Al-4V with ML models. Further studies determined the use of support vector machines (SVM) to forecast in AM, and the importance of data-driven techniques should be mentioned (Ref 29). Researchers have demonstrated the power of ML methods in the process parameters optimization and have pointed out the implementation of data-driven methods, making a particular emphasis on the involvement of ML in the enhancement of process control. The results demonstrate the importance of high-quality data as well as a coherent explanation of the model in order to improve the predictive model accuracy and reliability, and a way to a more efficient and performance-oriented manufacturing process (Ref 30, 31).

The use of ML regression models in attempting to predict mechanical and hardness behavior during metal 3D printing, especially of LPBF processes, is promising. The journals have published studies that reveal the effectiveness of different ML methods in attaining great prediction accuracy. Nonetheless, issues surrounding data quality, comprehensibility of models, and incorporation into manufacturing processes should be resolved. By specializing in ML, the upcoming studies will allow the metal 3D-printing industry to become more precise and efficient, which will eventually translate into the development of stronger products and more efficient production processes. The current article is an attempt to apply ML regression models in order to study the correlation between the LPBF process parameters and InconelTM 718 hardness. The other objective of this study is to determine the most appropriate hardness-testing method that will be used to accurately assess AM Inconel components.

2 Material and methods

2.1 Test Specimen

A design of experiment (DOE) method was devised to assess the hardness characteristics of InconelTM 718 printed via LPBF, according to (Ref 14). The parameters of the LPBF process affect the density of the material and, therefore, the properties of the material (Ref 32). The ED could be impacted by combining four primary parameters: layer thickness (s), laser power (P), hatch distance (h), and scanning speed (v) (Ref 33) (Fig. 1). ED, a parameter associated with material density, porosity, and the mechanism of forming the defects, measures the amount of energy that is given by unit volume of the powder placed. It has the following definition in Equation (1) and is measured in (J/mm³).

$$ED = \frac{P}{v \cdot s \cdot h} \quad (\text{Eq 1})$$

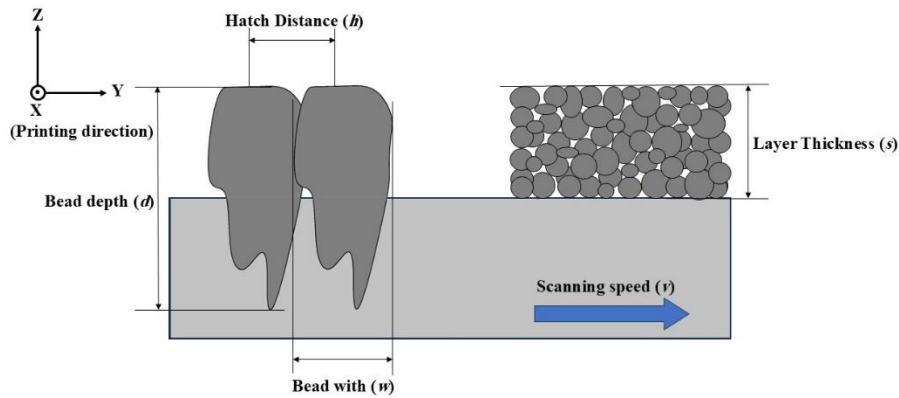


Fig. 1 The schematic representation of ED parameters

A similar parameter is suggested in (Ref 34), where an experimental investigation into the relationship between ED and surface roughness, pore size and number, and Vickers hardness is described. This parameter is specifically mentioned when the study addresses multiple process parameters concurrently. The test plan, described in (Ref 14), examines the effects of scanning speed and laser power. The arrangement of the specimens on the printing platform was designed to assess its impact on the material's hardness and response to seven distinct energy densities. Both deposition and growth planes were evaluated, and each specimen's hardness was measured. In (Ref 14), a detailed description of this experimental campaign is reported. To shorten the discussion, the reader is referred to the reference. Table 1 describes the specific process factors that regulate the internal and external geometry of the component. Additionally, Table 2 indicates the testing parameters, encompassing laser power (P), scanning speed (v), and ED.

Table 1: Main body and contour process parameters (Ref 14)

	Parameters	Values
Body	Layer thickness (mm)	0.03
	Spot Size (mm)	0.15
	Hatch distance (mm)	0.105
	Overlap Factor (mm)	0.7

Contour	Power (W)	192
	Spot size (mm)	0.15
	Beam compensation (mm)	0.075
	Contour Speed (mm/s)	1600
	Hatch Zone Border (mm)	0.065

Table 2: Body process parameters for different specimens (Ref 14)

Specimen Number	Laser Power (W)	Scanning Speed (mm/s)	Energy Density (J/mm ³)
1	192	400	152.38
2	192	600	101.59
3	192	800	76.19
4	230	500	146.03
5	230	600	121.69
6	230	700	104.31
7	230	900	81.13

2.2 Hardness Testing

The hardness was determined by an Innovatest Nemesis 9000 durometer, and linked with Impressions tester control and workflow software. This method is described and the results are presented in(Ref 14). The collection of data follows standard EN ISO 6507 (Ref 35, 36).

2.3 Dataset evaluation

Twenty eight cubical specimens printed in a Concept Laser M2 Series printer (Ref 37), using Inconel™ 718 powder (Ref 14) by Sophia High Tech (Napoli, Italy). The chosen powder, Powder Range 718, manufactured by Carpenter Additive (Philadelphia, PA, USA), was characterized in detail. Each of the specimens with a 20 mm edge was placed strategically on the fusion platform. The platform had four quadrants, which had seven specimens in each of the quadrants.

The hardness of Inconel™ 718 printed with LPBF was tested using a design of experiment (DOE) method, as shown in (Ref 14). Table 3 shows data on the hardness of a chosen subset of specimens in the as-built condition (before heat treatment) and after the heating treatment process. These measurements are made on the YZ /XZ planes and on the XY plane. In each test, the value of hardness is based on the average of five measurements in each surface of XY of each specimen and also the vertical surfaces (XZ/XY planes).

The complete data set represents that there are 56 observations, including the input and output features determined in the experimental trials. The input data include P , v , ED, test plane orientation (PO), and sample group. Output data will include the measures of hardness that will be taken right after the printing process (As Built) and after heat treatment (Heat Treatment). The two qualitative sets that constitute the input data can not be directly handled by any algorithms, but they play a large role in model complexity.

Table 3: Process parameters and measured hardness data of selected specimens (Ref 14)

Input Parameters				Hardness		
Laser Power	Scan Speed	ED	Plane	Sample No.	As Built	Heat Treatment
192	400	152.38	ZX/ZY	D	302	491
230	600	121.69	XY	B	291	474
192	800	76.19	XY	B	284	487
230	700	104.31	XY	D	292	475
230	600	121.69	XY	C	286	474
230	500	146.03	XY	A	284	474
230	700	104.31	XY	A	282	474
230	600	121.69	ZX/ZY	D	383	539
230	500	146.03	ZX/ZY	D	327	481
192	800	76.19	XY	A	287	485
230	700	104.31	ZX/ZY	A	316	478

2.4 Statistical Analysis and ANOVA

Regression and analysis of variance (ANOVA) represent fundamental statistical tools that are largely used in various research areas. They have tangible applications and useful objectives in industrial and commercial projects, particularly in mechanical fields that require improving the efficiency of the targeted output (Ref 38, 39, 40, 41, 42).

ANOVA is used to compare the variance contributions of sources and is beneficial to evaluate the effect of the factor that can be controlled on the research findings. The outcome of ANOVA, such as the P-value, is the most important result in that it determines whether the independent variables, namely printing parameters and sample orientation, have a significant effect on the measured feature, in the present case heat-treated hardness (Ref 43). The current study includes five different factors (N independent variables); therefore, an N-Way ANOVA was implemented.

2.5 Particle Swarm Optimization and Genetic Algorithm

Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) are specific in optimising parameters of predictive models, and design processes and system performances in areas such as mechanical engineering. PSO is used for solving complex optimization problems in a wide range of practical software's which is its widespread use in scientific and industrial approaches (Ref 44, 45, 46). The particles (solutions) were initialized randomly in the search space, then while changing each particle's direction (velocity), every particle holds its best value. Then the best value of all personal best values is taken as the global best value (swarm best value) (Ref 47). GA is a multi-path algorithm that searches many peaks in parallel, hence reducing the possibility of local minimum trapping and solving the multi-objective optimization problems (Ref 48). The GA process emphasizes the interplay between population, selection, crossover, and mutation (Ref 49, 50). This algorithm works iteratively by successively applying these operators in each generation till a termination criterion is satisfied (Ref 51).

Both PSO and the GA seek the correct global optimal solution, but the latter requires to spend more time for computations. The PSO is computationally more efficient as compared to GA because it has fewer function calls (Ref 52, 53). The advantages and drawbacks of the suggested optimization method were reviewed in (Ref 54, 55). In this section, a nonlinear relationship between parameters was considered. Although there are infinite possibilities of relationships

between input and output as expressed by deterministic expressions as an equation, this work will deal with single and two relationships between parameters. Equation (2) is introduced as the resultant formula.

$$\begin{aligned}
y = & b_1 x_1^{b_2} + b_3 x_2^{b_4} + b_5 x_3^{b_6} + b_7 x_4^{b_8} \\
& + b_9 x_5^{b_{10}} + b_{11}^{x_1} + b_{12}^{x_2} + b_{13}^{x_3} + b_{14}^{x_4} + b_{15}^{x_5} + b_{16} \left(\frac{x_1}{x_2} \right)^{b_{17}} \\
& + b_{18} \left(\frac{x_2}{x_3} \right)^{b_{19}} + b_{20} \left(\frac{x_3}{x_4} \right)^{b_{21}} + b_{22} \left(\frac{x_4}{x_5} \right)^{b_{23}} + b_{24} \left(\frac{x_1}{x_3} \right)^{b_{25}} \\
& + b_{26} \left(\frac{x_1}{x_4} \right)^{b_{27}} + b_{28} \left(\frac{x_1}{x_5} \right)^{b_{29}} + b_{30} \left(\frac{x_2}{x_4} \right)^{b_{31}} + b_{32} \left(\frac{x_2}{x_5} \right)^{b_{33}} \\
& + b_{34} \left(\frac{x_3}{x_5} \right)^{b_{35}} + b_{36} x_1 x_2 + b_{37} x_2 x_3 + b_{38} x_3 x_4 + b_{39} x_4 x_5 \\
& + b_{40} x_1 x_3 + b_{41} x_1 x_4 + b_{42} x_1 x_5 + b_{43} x_2 x_4 + b_{44} x_2 x_5 \\
& + b_{45} x_3 x_5
\end{aligned} \tag{Eq 2}$$

While in the current formula, b_1 to b_{45} are related to fixed equation coefficient that describes the overall relation between input parameters in 3D printing and the hardness of heat-treated samples as output and x_1 is related to P , x_2 is related to v , x_3 is related to ED, x_4 is related to PO and x_5 is related to sample number. In the above formula, all the parameters must be quantitative; therefore, a suitable conversion procedure is required for qualitative parameters like PO and sample number, as described in the results and discussion section.

2.6 Machine Learning

With recent progress, experimental data have started pouring in bulk, with large databases containing critical properties of materials. Such large databases, when judiciously used with the ML solution, hold great promise for improved processes in the manufacturing of materials and overall efficiency (Ref 56).

Support Vector Machines (SVMs) are well-known for their huge impact in addressing a variety of classification issues. The various salient points discussed about SVR are the Epsilon-Insensitive Zone, Function Approximation, Kernel Options, and Algorithm Sequence, all implemented in sequence to arrive at the final regression coefficient to refine the predictive methodology (Ref 57, 58).

Gaussian Process Regression (GPR) is an effective and non-parametric way of approaching regression problems and, as such, can be used in a case where we want to both explore and exploit. GPR in combination with Bayesian optimization and hyperparameter optimization is an opportunity to be able to interpret in probabilistic terms and take the direction of the most effective solution search (Ref 59). In addition, the GPR model, thanks to the flexibility of the kernel functions, allows accommodating a variety of problems and data types (Ref 60).

As the neural network (NN) learns, the weights and the strengths of these connections change. This learning process is referred to as training (Ref 61). The ANN is a very useful method of prediction. The ANN model is able to depend on input data to predict future outcomes effectively (Ref 62). ANN typically includes a single hidden layer and is suited for less complex projects. Deep ANNs enhance ANNs by integrating many hidden layers into deep architectures. In the same,

a summary of the advantages and disadvantages of the proposed optimization algorithm is reported (Ref 63, 64, 65).

Random Tree (RT) is a supervised ML algorithm that is applied in building predictive models, both for classification and regression problems. A random forest algorithm employs a random set of decision trees to calculate the optimal solution. A typical decision tree recursively subdivides the data. The objective at each division, starting from the root node, is to maximize the information gain achieved by that split (Ref 66, 67, 68).

Evaluation or error metrics, also known as performance metrics, are some of the essential operational tools in ML prediction models as well as regressions analysis. These metrics are a type of logical and mathematical tool. The aim of these tools is to assess the accuracy of predicted outcomes by comparing them to the actual results (Ref 69).

Because each prediction includes some error, these models must be measured to compare the performance of different ones and evaluate their (Ref 69, 70).

The Mean Squared Error (MSE), or more frequently called Mean Squared Deviation, is a measure of the squared difference between the observed and predicted values. It is a measure that assesses how well the line of best fit fits the observed data (Equation 3). A better or smaller MSE means a more accurate model. Root Mean Squared Error (RMSE) or Root Mean Squared Deviation (Equation 4) is a measure of the mean squared error. That is, RMSE is the standard deviation of the errors in the predictions, and it is a measure of how well the line of best fit closely approximates data which are observed data (Ref 71). Mean absolute error (MAE) measures the vertical difference (absolute value) between an actual value and its predicted value. Unless MSE, which shows the residuals' deviation, MAE provides a clear explanation of the average error in the same units as the target variable. This makes it easy to understand and communicate (see Equation 5) (Ref 72). Our performance measure of regression models is the Mean Absolute Percentage Error (MAPE), which is an intuitive relative error measure (Equation 6). It is particularly useful in assessing model, the usefulness of which is based on their ability to identify relatively large changes and not necessarily absolute deviations. However, MAPE can only work with positive values and inherently biases low forecasts, so MAPE is less applicable to models where it is expected that large errors are still possible (Ref 73).

The coefficient of determination, R^2 , is more complex when applied to nonlinear regression models or other nonlinear approaches, such as ANN-based predictions. In these cases, R^2 can fall outside the typical 0 to 100% range or even have negative values. Such a negative value indicates a poor fit of the model to the data (Ref 74). The literature discusses eight different formulas for R^2 . For nonlinear regression, where the model is not linear, a specific formula (Equation 7) is used to calculate the R^2 (Ref 75).

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_{pre} - Y_{act})^2 \quad (\text{Eq 3})$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_{pre} - Y_{act})^2} \quad (\text{Eq 4})$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_{pre} - Y_{act}| \quad (\text{Eq 5})$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \frac{|Y_{pre} - Y_{act}|}{Y_{act}} \quad (\text{Eq 6})$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_{pre} - \bar{Y}_{pre})^2}{\sum_{i=1}^n (Y_{act} - \bar{Y}_{act})^2} \quad (\text{Eq 7})$$

In the equations above, Y_{act} is the actual value of the output results from the dataset, Y_{pre} is the predicted value from any ML algorithm, $\bar{Y}_{act}$ is the mean value of the actual output, and $\bar{Y}_{pre}$ is the mean value of the predicted output.

Taylor diagrams are a kind of graphical tool used in many research studies in order to visualize the accuracy of models in targeted studies. This tool is mostly used in the visualization of a system, procedure, or process. The combination of three statistical metrics, including the centered root mean square difference (CRMSD), standard deviations (STD), and the Pearson correlation coefficient, is employed in this instrument. The data measures are put together in a single, easy-to-understand form. The duty of this tool is to enable the evaluation of the specific benefits of various models or the monitoring of changes in a model's performance. The graph shows that using the CRMSD metric, the model over-estimates the actual values. The reference point is assigned explicitly in the horizontal axis to be the ground truth. This shows the matching values for the real (Ref 76).

2.7 Structure of study

The current study consisted of three distinct parts, shown in Fig. 2. The first part (data preprocessing) involved reading data and converting qualitative inputs to quantitative (plane orientation and sample number). After preparing the quantitative data, the statistical analysis and ANOVA are implemented. The second part included model development and optimization algorithms (GA and PSO) to calculate the coefficient of the evaluated formula. Then, the formula by each method was implemented on the input data, and the results were compared with the experimental outputs from the dataset. The last part involved the implementation of five different ML approaches and measuring accuracy metrics for test and training data. However, before implementing ML, a random process divides the dataset into a train and a test set. All three parts had performance metrics and accuracy measurements, and the results were compared in order to find the best one for future predictions.

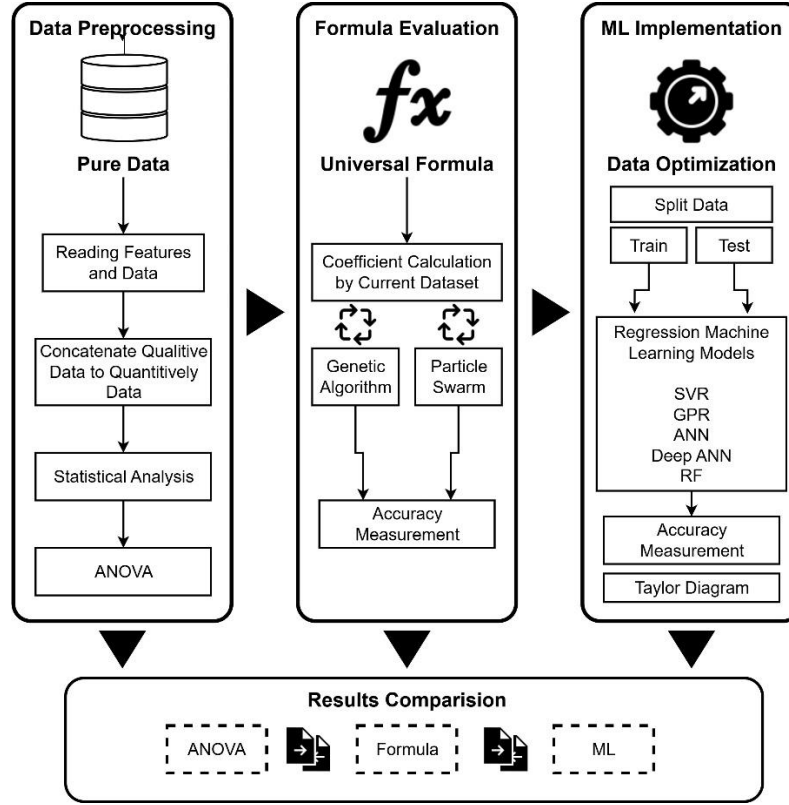


Fig. 2 The study procedure schema

3 Results and discussion

3.1 Statistical Analysis and ANOVA

The statistical analysis of the data record documented in Table 4 is a structured set of extraction and elaboration of mass data in an effort to unearth some trends and generate useful knowledge. Namely, the results outlined in Table 2 reveal that the domain range of any parameter should be considered in any further analytical or machine-learning activity. This kind of understanding supports the idea that one should remain aware of the distributional nature of variables when developing a model and undertaking tests. The reason is that the formulas and algorithms are valid only in this method within the current data range; however, predictions outside this range may lack accuracy.

Another important insight is the deviation of data, especially the output data. Since they require a greater variation within the output, indicated by a larger STD around the mean value, this makes the lower accuracy and stability, especially with ML algorithms. This means that under such scenarios, optimization would be required. Since this dataset shows a large variety from input to output data, and a large STD, the methodologies in the ML areas should be applied. It is the only reason that data normalization and optimization should be used during the ML process.

Table 4: Statistical Analysis of Current 56 Data in-out Dataset

	Min	Max	Average	STD
Laser Power	192	230	213.71	18.98
Scan Speed	400	900	642.85	160.52

Inputs	Energy Density	76.19	152.38	111.90	27.73
	Plane		Qualitative Data		
	Sample No.		Qualitative Data		
Output	Heat Treated	466	541	488.63	22.38

To explain the effects of various experimental conditions on the overall outcome of hardness, an ANOVA analysis (non-linear regression) was performed. In advance, they recoded the qualitative variables as plane and sample number into quantitative variables in order to get the chance of rigorous statistical manipulation. For the two planes orientation (XY and ZX/ZY), numbers 1 and 2, respectively, were assigned, while the four test samples (A, B, C, and D), numbers 1 to 4, were assigned. This preprocessed data set, now converted, serves as the main dataset for all implementations (Table 4).

The final results of the heat-treated hardness tests and the corresponding ANOVA are presented in Table 5. It is important to say that the primary output considered in all implementations is the heat-treated hardness. The heating process is carried out by implementing precise monitoring of heat treatment parameters, and the AM part properties can be made into optimized settings to suit the needs of its target application (Ref 77, 78).

Table 5: ANOVA Analysis Results

Parameter	Sum Sq.	d.f.	Mean Sq.	F	p-Value
Laser Power	443.6	1	443.62	1.38	0.2456
Scan Speed	4066.4	5	413.28	1.29	0.2852
Energy Density	0	0	0	0	NaN
Plane	9490	1	9490.02	29.61	0
Sample No	1113.5	3	317.16	1.16	0.3362
Error	14421.6	45	320.48		
Total	27535.1	55			

The ANOVA test conducted on the current data proved of no use in this research as far as the p-values of the test are concerned. When the p-value is less than 5% (which is equal to 0.05), it is easy to say that this type of variable dependency between input and output is rather considerable. The results, however, indicated that except for PO, all the parameters had a p-value over 5%. These results show that key parameters like LP, SS, or ED do not significantly affect the final hardness, which contradicts several prior investigations (Ref 79, 80, 81). So, the ANOVA test is unable to show the interdependence of all variables and needs deeper analysis. Complex techniques, including sophisticated mathematics and machine learning techniques, may be used in future research. Experimental techniques demonstrate that hardness is sensitive to the parameters confirmed by ANOVA; however, it fails to find a comprehensive link to the impact of all parameters on the final outcomes. Consequently, in this study, all parameters were selected for further investigation.

3.2 Formula Evaluation

Current research optimized the relationship between process parameters for Formula 1 and the final heat-treated hardness of products. In this section, a new formula that incorporates five different input variables was developed, shown in Equation 1. It includes five inputs: x_1, x_2, x_3, x_4 , and x_5 . The formula implementation concentrated on combinations of single and double inputs, yielding 45 coefficients. Although more complex combinations can be taken into account, this combination was enough. An optimization procedure using PSO and GA was used to estimate the coefficients, and this meant that the model captured the underlying relationships effectively.

PSO is intended to solve the optimization task by exploiting the objective function to specify the formula coefficients. MATLAB® software was used in this case. Equation 8 defines the objective function as the sum of squared errors between the predicted and actual values of hardness as drawn from the data. This is called an objective function where the goal of the optimization process is to make it as small as possible to get the optimal coefficients.

$$objective\ function = \sum_{all\ Data\ Set} (y_{formula} - y_{real})^2 \quad (Eq\ 8)$$

The algorithm iteration repeats the update steps until convergence criteria are met (a maximum number of iterations, which is defined as 100, or a satisfactory error threshold, which is defined as $1e^{-6}$). The hyperparameters of the current PSO implementation are tabulated in Table 6. The result of 10 iterations using MATLAB® has been obtained with the final formula being presented below in Equation 9 (the coefficients equal to zero have been dropped, and the formula has been simplified to make it easier to read).

Table 6: The hyperparameters of the current particle swarm implementation

Hyper Parameter	Value / Range	Description
Swarm Size	100	The number of particles in the swarm.
Number of Iterations	100	The maximum number of iterations or generations the algorithm will run
Inertia Weight (w)	[0.1, 1.1]	Controls the influence of the previous velocity on the current velocity
Cognitive Coefficient (c1)	1.5	The weight of the particle's own best position (personal best)
Social Coefficient (c2)	1.5	The weight of the global best position found by the swarm
Position Bounds	[0, 10]	The upper and lower bounds for the particle positions
Convergence Criteria*	$1e^{-6}$ / 1000	Conditions under which the algorithm will stop, such as a maximum number of iterations or a satisfactory error threshold
Random Numbers (r1 and r2) ¹	[0, 1]	Random numbers between 0 and 1 used in the velocity update equation

* Parameters that are automatically configured within the particle swarm algorithm.

$$\begin{aligned}
y = & 10x_2^{0.220} + 10x_3^{0.335} + 10x_5^{1.209} + 0.996x_1 + 0.761x_2 + 0.997x_3 + 6.463x_4 \\
& + 6.225x_5 + 10\left(\frac{x_2}{x_3}\right)^{0.166} + 10\left(\frac{x_3}{x_4}\right)^{0.358} + 10\left(\frac{x_1}{x_4}\right)^{0.074} \\
& + 10\left(\frac{x_2}{x_4}\right)^{0.389} + 10\left(\frac{x_2}{x_5}\right)^{0.124} + 10\left(\frac{x_3}{x_5}\right)^{0.457} + 50
\end{aligned} \tag{Eq 9}$$

This section addresses the optimization problem by minimizing the objective function by means of GAs to find the coefficients of the formula. The objective function computes the sum of squared errors between the predicted hardness (based on the computed coefficients) and the actual hardness values in the dataset, as specified by Equation 10. The update process was further carried out until convergence requirements were achieved either through 100 iterations or the error reduction reached below 1e-6. The hyperparameters of the presented GA implementation are given in Table 7. The last formula is given in Equation 10 and was determined after ten independent tests in MATLAB®. It is assumed that coefficients that have a zero value have been excluded and that, to make it easier to understand, the formula was simplified.

Table 7: The hyperparameters of the current GA implementation

Hyper Parameter	Value / Range	Description
Population Size	100	Determines the number of individuals in each generation. A larger population size can provide a more diverse set of solutions but may increase computational cost
Number of Generations	1000	Specifies how many iterations the algorithm will run. More generations can lead to better solutions but also require more computation time
Crossover Probability	0.8	The probability that two individuals will exchange parts of their chromosomes. This helps in combining good traits from different individuals
Mutation Function	@mutationpower	Specifies how mutations are applied to individuals
Selection Function	@selectiontournament	Determines how individuals are selected for reproduction
Elite Count	10	The number of top individuals that are guaranteed to survive to the next generation. This helps in preserving the best solutions found so far

$$\begin{aligned}
y = & 0.5x_1^{0.238} + 0.826x_2^{0.604} + 0.125x_3^{0.577} + 4.209x_4^{0.004} + 4.815x_5^{0.531} \\
& + 0.121x_1 + 0.999x_2 + 0.773x_3 + 0.843x_4 + 0.825x_5 \\
& + 0.421\left(\frac{x_1}{x_2}\right)^{9.079} + 1.075\left(\frac{x_2}{x_3}\right)^{1.730} + 3.423\left(\frac{x_4}{x_5}\right)^{0.039} \\
& + 1.423\left(\frac{x_1}{x_3}\right)^{0.296} + 5.176\left(\frac{x_1}{x_4}\right)^{0.051} + 0.168\left(\frac{x_1}{x_5}\right)^{1.030} \\
& + 0.003\left(\frac{x_2}{x_4}\right)^{1.325} + 0.822\left(\frac{x_2}{x_5}\right)^{0.516} + 0.930\left(\frac{x_3}{x_5}\right)^{0.967} \\
& + 0.031x_3x_4 + 0.004x_4x_5 + 0.006x_1x_3 + 0.004x_1x_4 \\
& + 0.131x_1x_5 + 0.056x_2x_4 + 0.074x_3x_5
\end{aligned} \tag{Eq 10}$$

The overall results formula from PSO and GA optimization method implemented on input data and formula output compared one by one with real values, and the accuracy metrics implemented on each series of data (Real Value and Formula Value), the overall results are tabulated in Table 8. As shown in Table 8, the PSO provided a better solution due to the lower MSE and MAE, and also the mean accuracy for all the given 56 data points is around 97% with respect to 93% for GA. Furthermore, the generated formula by PSO is also simpler than GA, which aligns with the advantages of PSO with respect to GA mentioned in Table 8.

Table 8: Accuracy metrics for evaluated form formula implemented by PSO and GA

Accuracy Metrics	Particle Swarm	Genetic Algorithm
Accuracy (Max)	100%	100%
Accuracy (Min)	92%	82%
Accuracy (Mean)	97%	93%
MSE	349.2	1806.2
RMSE	18.7	42.5
MAE	15.3	35.4
MAPE	3.1%	7.2%
R ²	0.29	-2.7

However, for both solutions, the final result is a single fixed formula that is sensitive to the input domain. Both obtained formulas (Equations 2 and 3) have the following constraints:

- x_1 the laser power should be in the range of 192 to 230
- x_2 the scan speed should be in the range of 400 to 900
- x_3 the energy density should be in the range of 76.19 to 152.38
- x_4 the plane orientation should be selected from an integer number between 1 and 2
- x_5 the sample number should be selected from an integer number between 1 to 4

These limitations restrict the use case of this formula when used universally, as it only holds true in the existing data. Also, the implementation cannot adapt to any other data domains, and it lacks scalability. In case the set of data is changed or more parameters are introduced, the formula would not be useful anymore with the new observations. Therefore, enhancing new methods to solve these limitations to obtain flexibility and scalability is necessary.

3.3 Machine Learning Implementation

To apply the five different ML algorithms, the dataset was divided into two groups. The first group is for the training process in the ML methods, which had a large amount of data and, for this study, includes 45 samples of data; the remaining data includes 11 sets of data, which were used for validating the algorithms and called testing data.

The property of each group (train and test) is mentioned in Table 9. The train and test datasets exhibited distinct statistical actuality to confirm the efficacy of the data splitting technique and to guarantee an acceptable distribution of the data, but as mentioned before, both of them have the same domain, which is crucial in the realm of ML.

Table 9: Statistical property of train and test dataset

Variable name		Min		Max		Average	
		Train	Test	Train	Test	Train	Test
Input	Laser Power	192	192	230	230	213.1	291.6
	Scan Speed	400	400	900	800	653.3	627.3
	Energy Density	76.19	76.19	152.38	152.38	109.9	115.9
	Plane	1	1	2	2	1.6	1.4
	Sample No.	1	1	4	4	2.5	2.5
Output	Heat Treated	466	474	541	539	489.7	484.7

In the process of analyzing, there are hyperparameters that are considered for ML algorithms. The ML models will operate well if their hyperparameters are chosen accurately and properly. In this study, the Bayesian optimization method was employed in order to find the best hyperparameters. MATLAB® software with its standard rules was employed to operate the process of optimizing the variables. The final results are presented in Table 10.

As mentioned above, Bayesian optimization was employed to find the ideal hyperparameters. Speaking of the standard Bayesian optimization, with five-fold cross-validation was used in the process. Because every model had many hyperparameters, in this study, several important hyperparameters were considered, and the remaining were default values in the software package. Three parameters, such as the regularization parameter (C), epsilon, and the type of kernel function, were optimized for using the SVR model. The C parameter is used for the level of penalty applied to data points that are outside the particular region. The epsilon parameter gives a range of tolerance of the error of the regression line, outside which the deviations need not be deemed as errors. The flexibility of the fitting function is controlled by the type of kernel function. For the GPR model, the sigma parameter of the squared exponential kernel and the type of kernel covariance function were optimized. The kernel covariance function could be chosen from a variety of options: rational quadratic, squared exponential, Matern kernel with parameter 5/2, and exponential. These options defined the flexibility of the model to better fit the input data. For the ANN and Deep ANN models, the number of hidden layers, the total number of neurons in each hidden layer, the activation function, and the learning rate were optimized. The difference between standard and deep neural networks is the size of the hidden layer, where for the standard one, just one hidden layer is considered, while for the deep model, more than two hidden layers are considered. The number of neurons in each layer seemed to improve the performance of the ANN model, but it also increased the risk of overfitting. An activation function is crucial for determining how the network processes and outputs data. There are several choices for activation functions like sigmoid, Relu, and tanh, but the Relu activation function is a linear function that is suitable for

linear and nonlinear problems. For RT, the smaller leaf sizes can capture more detail but may lead to overfitting. A maximum number of splits will control the maximum number of decision splits in the tree. Greater splits may make any given model more complex. Split Criterion specifies the criteria on which the nodes are split based on, e.g., mean squared error (MSE). To address the issue of missing data, one takes advantage of missing data through the use of Surrogate Decision Splits, which seek alternative splits. Pruning involves removing branches that have little importance to reduce overfitting.

Table 10: The all-hyper-parameters used in the implementation

ML Models	Hyperparameters	Search Space	Selected Value
SVR	C	(0.1, 1000)	13.89
	Epsilon	(0.01, 10)	1.39
	Kernel function	polynomial, gaussian, linear	Polynomial degree 3
GPR	Sigma	(0.01, 10)	4.5
	Kernel Parameter	(0.01, 10)	6.5
	Kernel function	RQ, SE, M52, Exp.	RQ
ANN	Neurons in each layer	(5, 20)	5
	Learning rate	(0.01, 1)	0.1
	Activation Function	Relu, Sigmoid, Tanh	Relu
Deep ANN	Hidden layers	(2, 10)	3
	Neurons in each layer	(5, 20)	[8, 8, 8]
	Learning rate	(0.01, 1)	0.1
	Activation Function	Relu, Sigmoid, Tanh	Relu
RT	Minimum Leaf Size	(1, 20)	5
	Maximum Number of Splits	(10, 100)	50
	Split Criterion	MSE	MSE
	Surrogate Decision Splits	(on, off)	on

After performing up to 100 iterations for the final selection of each hyperparameter and comparing the accuracy metrics values as the verification factors, every algorithm was trained on 45 data points. As shown in Table 11, ML implementation does not provide acceptable results for finding the best solution, and therefore final data distribution around the perfect prediction line is not sufficient. Although the final absolute mean error (MAPE) was so low, under 6% for all five methods, the mean absolute accuracy for all five methods is more than 94%. However, because of the large amount of data deviation, which is shown in the R^2 parameter, these five implementations are not acceptable for any future implementation. Therefore, these methods are not stable in the training process. The optimization process happened at two distinct levels, the first one happens at the hyperparameters side, which is implemented and reaches acceptable accuracy, but not sufficient for method stability, so the other side is at the data preparation to reach a better solution.

Table 11: Accuracy metrics for training process non-optimized parameters

Data set	Accuracy Metrics	SVR	GPR	ANN	Deep ANN	RT
Train	MSE	886.0	483.5	461.2	1658.1	522.1
	RMSE	29.8	22.0	21.5	40.7	22.8
	MAE	21.7	17.5	16.8	29.5	16.7
	MAPE	4.3	3.5	3.4	5.9	3.3
	R ²	-0.68	0.08	0.12	-2.20	0.01

An essential process of an ML is feature engineering, accepting a feature vector as an input and delivering a prediction. These are supervised trained models, and in these models, the feature vectors that are presented correspond to the actual expected output. Some features may be improved through approaches like normalisation, logarithmic conversion, and formalisation of new features with polynomials that make the data more suitable for being worked by linear models. Moreover, developing additional characteristics, such as interaction terms, and identifying the most relevant ones via various approaches may substantially improve model performance and stability (Ref 82, 83).

Several methods are proposed on feature engineering, in the section of feature construction, combining input and output data with specific formulas to capture their interaction effects that enhance the stability of ML implementation. In the following section, to reach a better solution and a stable ML algorithm, the new output is based on the combination of inputs and outputs proposed by Equation 11. Although this equation is based on the fixed input data, and therefore the final results can be easily converted to the main hardness value, the flow is able to roll back, which is important to reach a final reasonable solution.

Hardness_{optimized}

$$= \text{Hardness}_{HT} + LP + \left(6.43 - \frac{SS}{100}\right) - ED + 20^{Plane} + 100 * \text{Sample Number} \quad (\text{Eq 11})$$

The optimized training results based on the two mentioned optimized approaches (hyperparameter optimization and feature engineering based on Equation 11) are illustrated in Fig. 3 (a) to (e). These Figures show the correlation between predicted results during the process and real hardness from the perfect prediction line. The accuracy metrics for the training process optimized parameters are tabulated in Table 12. As illustrated results, the GPR and single-layer ANN proposed the best training results. The R² for both methods is more than 0.98 (0.99 for GPR and 0.98 for ANN), which is perfect in the ML domain. The formula changes the domain of data and now new output data have the minimum and maximum values of 639.34 and 1448.53 respectively with a mean value of 1070.3 which affects the other accuracy metrics parameters, and it is impossible to compare MSE and RMSE before and after feature engineering, but the MAPE parameter indicate a mean absolute error in percentage and also it is comparable. The MAPE did not change reasonably and remained under 3.5% and 3.4% for GPR and ANN, respectively, and indicates that both optimization approaches provided the best solution by these methods.

The SVR method reaches a good R² and is more than 0.91. Furthermore, the absolute mean percentage error increases from 4.3% to 4.7%, but the total mean accuracy remains around 95%, which is also acceptable in the engineering domain.

Although the Deep ANN provided complex solutions in some areas, it did not do so in others. Like the other method, the optimization process improved the Deep ANN, and R^2 reached the perfect result of 0.96. Furthermore, the MAPE decreased from 5.9% to 3%. The last method, RT, also just reached an R^2 of about 0.88 and a MAPE under 7%.

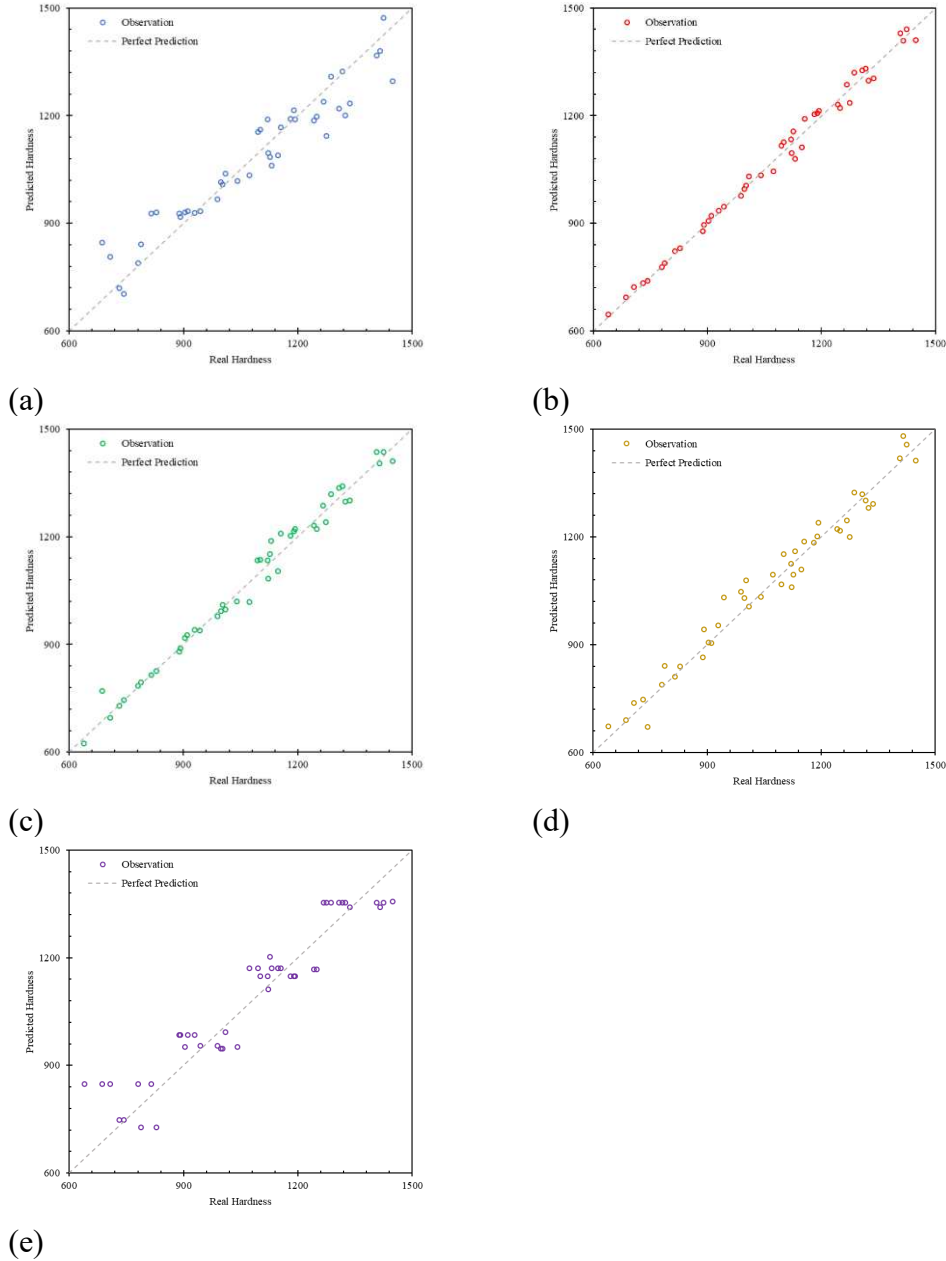


Fig. 3 Optimized training results (a) SVR, (b) GPR, (c) ANN, (d) Deep ANN, (e) RT

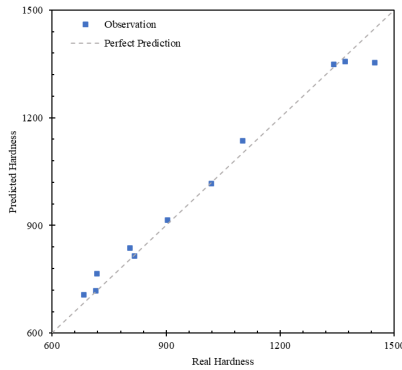
The overall results from the current section show that all the methods, even RT with the lowest R^2 related to others, have sufficient stability and accuracy (more than 93%), and for continuing investigation and implementation on testing data is suitable; however noted that testing data should be optimized output like a training dataset for implementing testing procedure.

Table 12: Accuracy metrics for training process optimized parameters

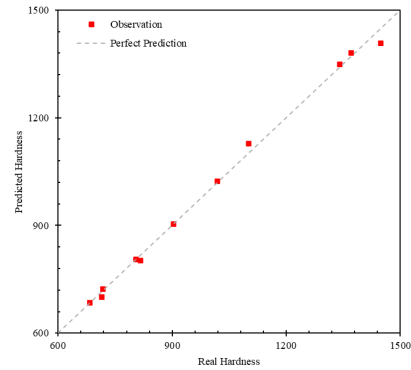
Data set	Accuracy Metrics	SVR	GPR	ANN	Deep ANN	RT
Train	MSE	3910.4	416.9	794.1	1505.2	5663.2
	RMSE	62.5	20.4	28.2	38.8	75.3
	MAE	47.8	16.4	21.9	31.5	62.4
	MAPE	4.7	1.5	2.1	3.0	6.5
	R^2	0.91	0.99	0.98	0.96	0.88

After training data, each method reached stable and reasonable parameters within its algorithm. It is time to test the models on test data. Test data is a small portion of the main data set, which is completely split from the training data set and includes data within the domain of the training data set. Therefore, the final results of testing these 13 sets of data are illustrated in Fig. 4 (a) to (e), for SVR, GPR, single-layer ANN, Deep ANN, and RT, respectively. Furthermore, the accuracy metrics which is associated with the final testing results are tabulated in Table 13.

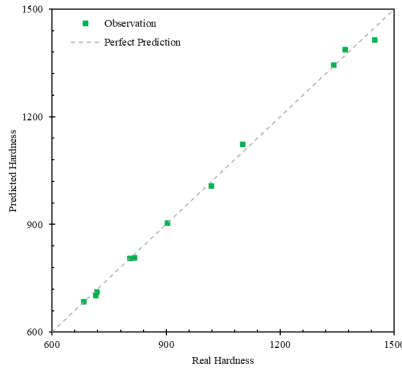
As a result of the training process, the GPR and single-layer ANN bring sophisticated results with respect to the other methods. Again, in the testing procedure, both of them had a perfect R^2 , which is 0.99, and the lowest MAPE (1.3 for both GPR and ANN), so these methods were selected among them as the best ML implementation.



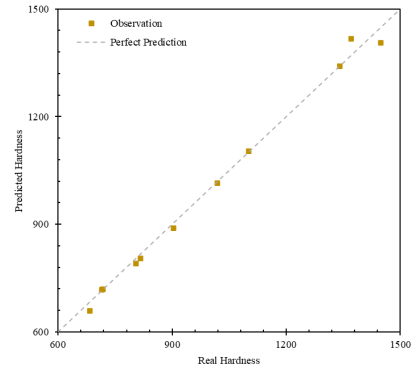
(a)



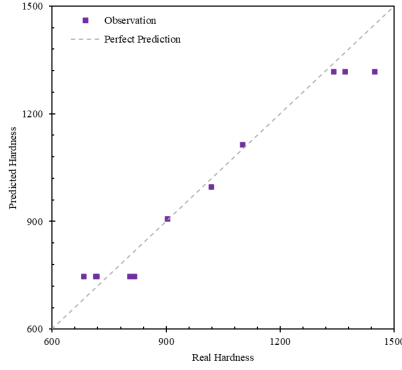
(b)



(c)



(d)



(e)

Fig. 4 Optimized testing results (a) SVR, (b) GPR, (c) ANN, (d) Deep ANN, (e) RT

At the next level, the SVR and Deep ANN with R^2 of more than 0.98 and MAPE of less than 2.6% provided acceptable results, but again RT method has the lowest R^2 of about 0.95 and the highest mean absolute percentage error of about 4.6% like training results, provided worst results among mentioned ML algorithm (Table 13).

Table 13: Accuracy metrics for testing process optimized parameters

Data set	Accuracy Metrics	SVR	GPR	ANN	Deep ANN	RT
Test	MSE	1330.2	313.5	260.5	483.6	3211.2
	RMSE	36.5	17.7	16.1	22.0	56.7
	MAE	26.1	13.6	13.3	15.4	44.2
	MAPE	2.6	1.3	1.3	1.4	4.6
	R^2	0.98	0.99	0.99	0.99	0.95

The process of optimization is based on feature engineering and output data is changed to a new version which is calculated by equation 4, but the final output which is measured by the testing process should be calculated by the rollback process, the rollback process means calculated the final predicted hardness from newly generated parameter in the testing process by equation 5 which is reversed version of equation 12.

$$\begin{aligned}
 Hardness_{Predicted} &= Hardness_{Optimized\ Test} \\
 &- \left(LP + \left(6.43 - \frac{SS}{100} \right) - ED + 20^{Plane} + 100 \right. \\
 &\quad \left. * Sample\ Number \right)
 \end{aligned} \tag{Eq 12}$$

In the following formula $Hardness_{Optimized\ Test}$ is generated from the testing process and then put in equation 12 with related input including LP, SS, ED, PO, and Sample Number and because these input parameters for each set of data is fixed the rollback process act like a linear formula and the $Hardness_{Predicted}$ generated from this method is optimized predicted hardness which is compared with both real hardness and non-optimized predicted for all five method; the overall results mentioned in the Fig. 5 (a) to (e) and also Table 14.

As mentioned before, the SVR method did not bring the best results in both the training and testing processes related to MAPE and R^2 ; However, the accuracy remains acceptable at more than 95%. Based on the results in Table 14 for RT, the mean accuracy for both optimized rollback and non-optimized models is still high. The MAPE parameter also shows the mean absolute error, which is not more than around 5%.

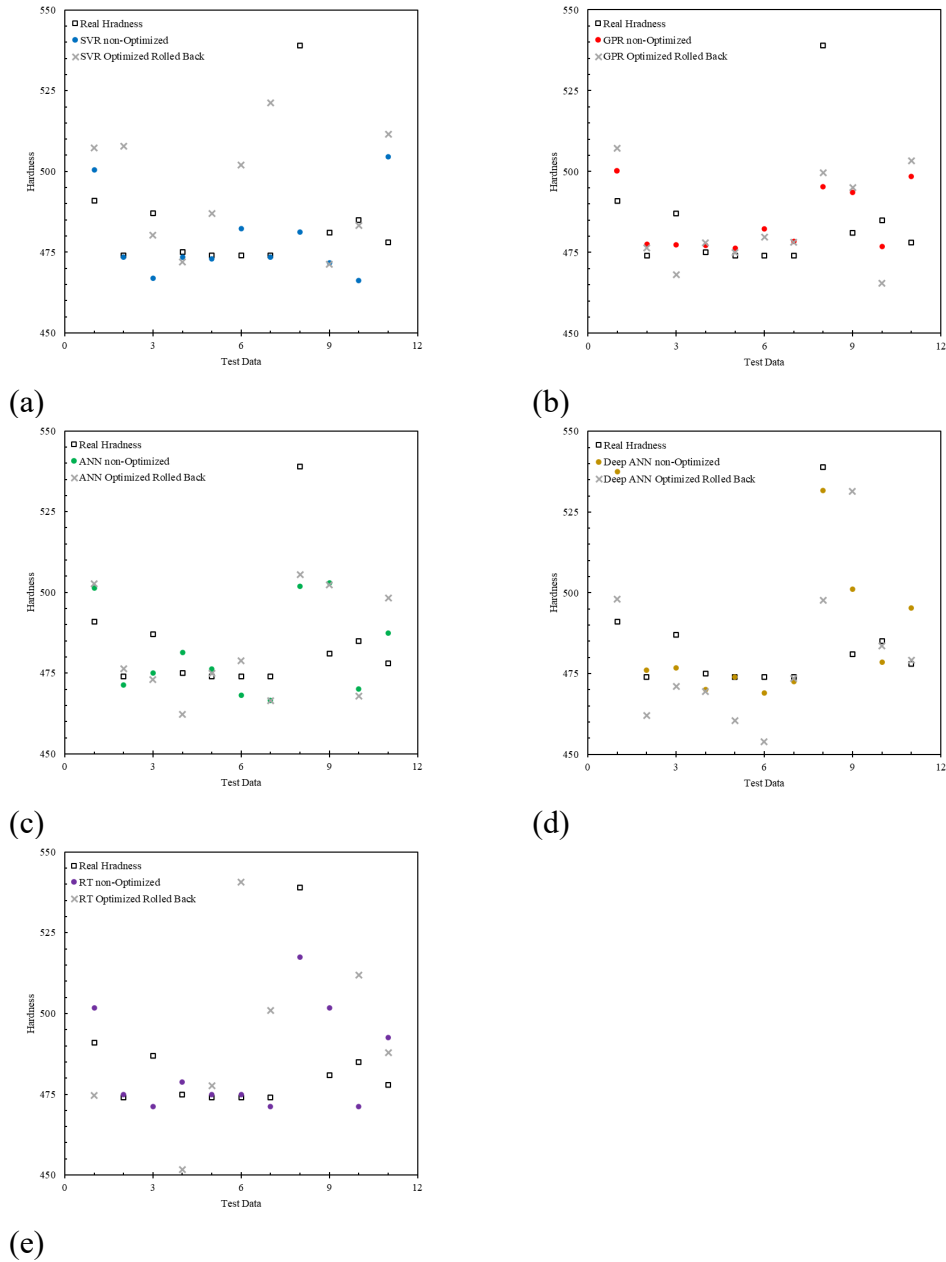


Fig. 5 Comparing experimental hardness with non-optimized and optimized rollback by (a) SVR, (b) GPR, (c) ANN, (d) Deep ANN, (e) RT methods

Table 14: Results of hardness with non-optimized and optimized rollback for different methods

HT	SVR	SVR Roll Back	GPR	GPR Roll Back	ANN	ANN Roll Back	Deep ANN	Deep Roll Back	RT	RT Roll Back
491	500.4	507.2	500.1	507.2	501.3	502.7	537.4	498.1	501.7	474.6
474	473.4	507.8	477.5	476.4	471.3	476.4	476.0	462.0	475.0	417.3
487	466.9	480.2	477.3	468.2	475.1	473.1	476.7	471.1	471.2	411.8
475	473.5	471.9	477.3	478.0	481.4	462.2	470.0	469.4	478.8	451.6
474	473.0	487.1	476.3	475.0	476.3	475.0	473.9	460.4	475.0	477.6
474	482.3	502.0	482.3	479.8	468.2	478.9	469.0	453.9	475.0	540.7
474	473.5	521.3	478.4	478.2	466.6	466.4	472.6	473.6	471.2	501.0
539	481.3	445.4	495.4	499.7	502.0	505.5	531.7	497.6	517.4	407.9
481	471.6	471.3	493.6	495.1	503.0	502.4	501.1	531.4	501.7	431.3
485	466.2	483.3	476.9	465.5	470.2	467.9	478.6	483.6	471.2	511.8
478	504.5	511.5	498.5	503.3	487.3	498.3	495.4	479.1	492.5	487.8
Mean Accuracy	97%	95%	98%	97%	98%	97%	98%	97%	98%	91%
MAE	14.0	26.1	11.3	13.6	12.2	13.3	11.0	15.4	9.7	44.2
MAPE	2.8	5.3	2.3	2.8	2.5	2.7	2.3	3.1	2.0	8.9

On the other hand, the GPR method brings the best results for the training and testing process, and is one of the selection candidates for the proposed study. In Table 14 for GPR and Fig. 5 (b), the overall results of GPR rollback are also acceptable and bring the absolute mean percentage error around 2.8% and mean accuracy around 97% which is similar to 98% accuracy non-optimized process and less than 1% change in accuracy is absolutely neglectable in the field of mechanical engineering.

Like GPR, the single-layer ANN with 5 hidden layers and Relu activation function also brings the best results for both the training and testing process. The comparison between a non-optimized ANN which is mentioned in the column of ANN in Table 14 with the optimized rolled back shows that both have the same mean absolute error (2.5% for non-optimized and 2.7 for optimized rollback) and it is clear that both methods don't have any difference, and the optimized process acts perfectly on model stability. Therefore, this ANN method was selected as the best implementation in the current study.

In this study, the Deep ANN implementation did not bring the best results and was chosen after the ANN and GPR methods at the third stage. However, by comparing the final hardness results from the prediction method both in the non-optimized process and optimized rollback process which is mentioned in Table 14 for Deep ANN, the accuracy remains acceptable and more than 97% and the mean absolute error is not more than around 3%, and also the accuracy difference between optimized and non-optimized around 1% which is reasonably acceptable.

The last implemented method is RT, which has the worst results among these five ML methods, both in the training and testing process, in an optimized manner. From Table 14 for RT, the optimized rollback also brings the mean absolute percentage error around 9%, which is increased by 7% compared mean absolute percentage error by the non-optimized process and absolutely not acceptable in the current study.

3.4 Taylor diagrams

The STD, Pearson correlation, and centered RMSD are the three important metrics that are used in the Taylor diagram, which is used to compare the performance of different machine learning models. These metrics determine and show the degree of similarity between the predictions generated by ML models and the outcomes of experiments. In cases where one ML model performs successfully, this will be due to a low RMSD, following a high correlation. Also, its standard deviation closely matches the experimental outcomes. Therefore, the targeted model will be positioned closer to the experimental point on the graph. This would ensure its higher accuracy in comparison with other models. From the Taylor diagram (Fig. 6), the GPR and ANN methods provided the best solution, and again, this method was suggested as the best method during the entire process.

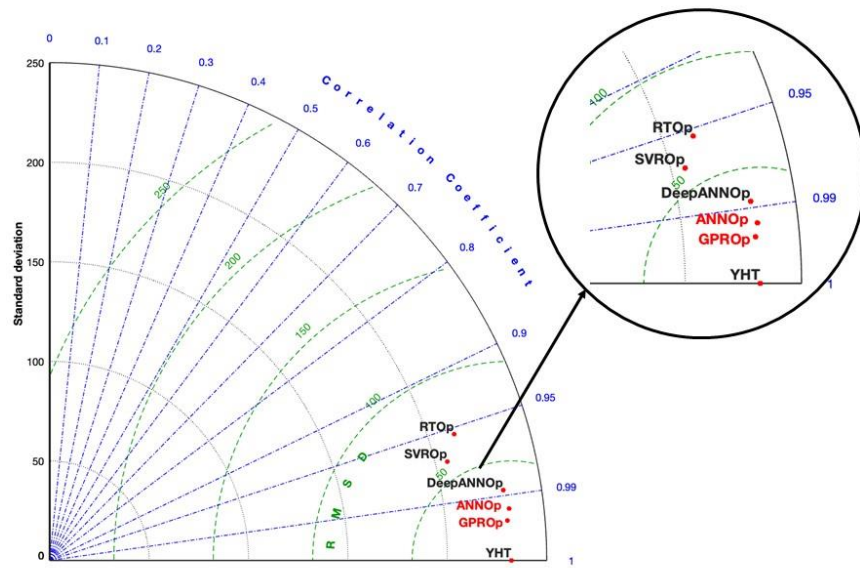


Fig. 6 Taylor diagram for all optimized ML method

Conclusions

This paper focuses on the evaluation of the impact of the printer setting parameters on the ultimate product hardness of the LPBF technique on the M2 Series 5 3D printer model by General Electric Additive Solutions with Inconel™ 718 as a material. Laser power, scan speed, energy density, orientation of test plane, and sample group were used as the prime input parameters. The hardness measurements were obtained both before and after the treatment of heat in two planes and on 28 particles.

In the initial step, statistical analysis was being done using the technique called ANOVA regression, where the data was preprocessed so as to convert qualitative data (PO and Sample Group Number) to make it quantitative data. The ANOVA tests, however, show that there is no significant influence of the inputs on outputs.

In the second stage, a sophisticated formula was developed based on input and output data. These data are single and sequential relations among five parameters, having 35 different coefficients. This formula was solved through two optimization techniques, namely PSO and GA. Although the results obtained possessed low mean absolute percentage errors (3.1% for PSO and 7.2% for GA), the data deviation was too great, and the values of R^2 were under 0.3. So, this indicates poor applicability for general use and inflexibility with new data or parameters.

In order to overcome these drawbacks, ML techniques were taken into account. They included five regression algorithms: SVM, GPR, single-layer ANN, Deep ANN, and RT. First of all, the data were divided into 45 training data points and 11 testing data points. Optimization of the hyperparameters was done using the Bayesian algorithm, but the R^2 values were low or negative, indicating that it was not stable.

Feature engineering was introduced to combine parameters and define new features. This creates a sophisticated solution for all five ML algorithms. The solution provided R^2 greater than 0.91 in SVM, GPR, ANN, and Deep ANN, and 0.88 in RT, with the mean accuracy of approximately 93% and a maximum MAPE of 6.5%. GPR and single-layer ANN showed R^2 values of 0.99 and 0.98 and MAPE values of 1.5% and 2.1%, respectively, as the best training results.

The maximum R^2 (0.99) values and MAPE (1.3%) were attained on the testing data by GPR and ANN, which is confirmation of the two models being the optimal solutions. There was a rollback process of comparing the predicted optimized hardness against the actual hardness. This shows that GPR and ANN are reliable since they perform well and attain an optimized mean accuracy of about 97.

The study concludes that the ML methods, GPR and ANN, are very proficient in predicting hardness in LPBF processes. This shows dependence on the parameter quantity and its future employment in other additional applications, such as online deployment. Future research will focus more on improving prediction accuracy by treating coolant and dresser as independent parameters. The next study to automate the process of smart machining to reach the desired levels of hardness would be to use digitized data through the Industrial Internet of Things (IIoT).

Acknowledgments

This publication is also part of the project PNRR-NGEU which has received funding from the MUR-DM117/2023 for one of the writers (MD), [Project No: MUR DM 117/Tsubaki Nakashima – Recovery, reuse and reduction of waste sludge in the production of balls and rolls for rolling bearings] (Nov. 2023- Oct. 2026), which is gratefully acknowledged. The Authors gratefully thank Sophia High Tech (Napoli, Italy) for providing samples for experimental evaluation and for their continuous support in research activities

References

1. X. Zhang, R. Tiwari, A.H. Shooshtari, and M.M. Ohadi, An additively manufactured metallic manifold-microchannel heat exchanger for high temperature applications, *Applied Thermal Engineering*, 2018, 143, p 899-908
2. I. Yadroitsev, I. Yadroitsava, A. Du Plessis, and E. MacDonald, Fundamentals of laser powder bed fusion of metals, Elsevier, 2021
3. Y. Kok, X.P. Tan, P. Wang, M. Nai, N.H. Loh, E. Liu, and S.B. Tor, Anisotropy and heterogeneity of microstructure and mechanical properties in metal additive manufacturing: A critical review, *Materials & Design*, 2018, 139, p 565-586

4. P.R. Gradl, S.E. Greene, C. Protz, B. Bullard, J. Buzzell, C. Garcia, J. Wood, R. Osborne, J. Hulka, and K.G. Cooper, Additive manufacturing of liquid rocket engine combustion devices: a summary of process developments and hot-fire testing results, 2018 Joint propulsion conference, 2018, p 4625
5. S. Soller, R. Behr, S. Beyer, F. Laithier, M. Lehmann, A. Preuss, and R. Salapete, Design and testing of liquid propellant injectors for additive manufacturing, 7th European Conference for Aeronautics and Aerospace Sciences, EUCASS), Milano, Italy, 2017
6. D. Liuzzi, C. Boffa, M. Rudnykh, D. Drigo, L. Arione, N. Ierardo, and A. Sirbi, Development of an ALM thrust chamber for Vega-E M10 rocket engine, *Space Propuls*, 2021, 2020(1), p 1-8
7. M. Balbaa, S. Mekhiel, M. Elbestawi, and J. McIsaac, On selective laser melting of Inconel 718: Densification, surface roughness, and residual stresses, *Materials & Design*, 2020, 193, p 108818
8. B. Blakey-Milner, P. Gradl, G. Snedden, M. Brooks, J. Pitot, E. Lopez, M. Leary, F. Berto, and A. Du Plessis, Metal additive manufacturing in aerospace: A review, *Materials & Design*, 2021, 209, p 110008
9. K. Moussaoui, W. Rubio, M. Mousseigne, T. Sultan, and F. Rezai, Effects of Selective Laser Melting additive manufacturing parameters of Inconel 718 on porosity, microstructure and mechanical properties, *Materials Science and Engineering: A*, 2018, 735, p 182-190
10. Y. Li, M. Založnik, J. Zollinger, L. Dembinski, and A. Mathieu, Effects of the powder, laser parameters and surface conditions on the molten pool formation in the selective laser melting of IN718, *Journal of Materials Processing Technology*, 2021, 289, p 116930
11. W.D. Callister and D.G. Rethwisch, Fundamentals of materials science and engineering, Wiley London, 2000
12. R.G. Budynas and J.K. Nisbett, Shigley's mechanical engineering design, McGraw-Hill New York, 2011
13. E. Pavlina and C. Van Tyne, Correlation of yield strength and tensile strength with hardness for steels, *Journal of materials engineering and performance*, 2008, 17, p 888-893
14. R. Sesana, C. Delprete, M. Pizzarelli, M. Crachi, L. Lavagna, D. Borrelli, and A. Caraviello, Influence of Deposition Parameters on Hardness Properties of InconelTM 718 Processed by Laser Powder Bed Fusion for Space Applications, *Journal of Manufacturing and Materials Processing*, 2023, 7(1), p 36
15. V. Dovale-Farelo, P. Tavadze, L. Lang, A. Bautista-Hernandez, and A.H. Romero, Vickers hardness prediction from machine learning methods, *Scientific Reports*, 2022, 12(1), p 22475
16. S. Nasiri and M.R. Khosravani, Machine learning in predicting mechanical behavior of additively manufactured parts, *Journal of materials research and technology*, 2021, 14, p 1137-1153
17. P. Charalampous, N. Kladovasilakis, I. Kostavelis, K. Tsongas, D. Tzetzis, and D. Tzovaras, Machine learning-based mechanical behavior optimization of 3D print constructs manufactured via the FFF process, *Journal of Materials Engineering and Performance*, 2022, 31(6), p 4697-4706

18. S. Dong, Y. Wang, J. Li, Y. Li, L. Wang, and J. Zhang, Machine Learning Aided Prediction and Design for the Mechanical Properties of Magnesium Alloys, *Metals and Materials International*, 2024, 30(3), p 593-606
19. W. Zhu, W. Huo, S. Wang, Ł. Kurpaska, F. Fang, S. Papanikolaou, H.S. Kim, and J. Jiang, Machine learning-based hardness prediction of high-entropy alloys for laser additive manufacturing, *JOM*, 2023, 75(12), p 5537-5548
20. C.L. Mambuscay, C. Ortega-Portilla, J.F. Piamba, and M.G. Forero, Predictive Modeling of Vickers Hardness Using Machine Learning Techniques on D2 Steel with Various Treatments, *Materials*, 2024, 17(10), p 2235
21. U.M.R. Paturi, S.T. Palakurthy, S. Cheruku, B. Vidhya Darshini, and N. Reddy, Role of machine learning in additive manufacturing of titanium alloys—a review, *Archives of Computational Methods in Engineering*, 2023, 30(8), p 5053-5069
22. J. Xiong, T. Zhang, and S. Shi, Machine learning of mechanical properties of steels, *Science China Technological Sciences*, 2020, 63(7), p 1247-1255
23. P. Akbari, M. Zamani, and A. Mostafaei, Machine learning prediction of mechanical properties in metal additive manufacturing, *Additive Manufacturing*, 2024, 91, p 104320
24. A. Stoll and P. Benner, Machine learning for material characterization with an application for predicting mechanical properties, *GAMM-Mitteilungen*, 2021, 44(1), p e202100003
25. Z. Gu, S. Sharma, D.A. Riley, M.V. Pantawane, S.S. Joshi, S. Fu, and N.B. Dahotre, A universal predictor-based machine learning model for optimal process maps in laser powder bed fusion process, *Journal of Intelligent Manufacturing*, 2023, 34(8), p 3341-3363
26. A. Deka and J. Hall, Predictive modeling of mechanical properties for fused deposition modeling parts: A focus on processing and environmental parameters, *Manufacturing Letters*, 2023, 35, p 643-651
27. S.D. Grozav, A.D. Sterca, M. Kočiško, M. Pollák, and V. Ceclan, Artificial neural network-based predictive model for finite element analysis of additive-manufactured components, *Machines*, 2023, 11(5), p 547
28. Q. Luo, J.D. Shimanek, T.W. Simpson, and A.M. Beese, An image-based transfer learning approach for using in situ processing data to predict laser powder bed fusion additively manufactured Ti-6Al-4V mechanical properties, *3D Printing and Additive Manufacturing*, 2024,
29. F. Yan, Y.-C. Chan, A. Saboo, J. Shah, G.B. Olson, and W. Chen, Data-driven prediction of mechanical properties in support of rapid certification of additively manufactured alloys, *Computer Modeling in Engineering & Sciences*, 2018, 117(3), p 343-366
30. M. Banerjee, A. Banerjee, D. Mukherjee, A. Singla, and J. Singh, Machine Learning Module for Predicting Tensile Response of SLMed Ti-6Al-4V, *Advances in Additive Manufacturing and Metal Joining: Proceedings of AIMTDR 2021*, Springer, 2023, p Page
31. V. Maitra and J. Shi, Surface Roughness Prediction for Additively Manufactured Ti-6Al-4V Components Based on Supervised Learning Models, *International Manufacturing Science and Engineering Conference*, American Society of Mechanical Engineers, 2022, p V001T001A013
32. L. Tonelli, A. Fortunato, and L. Ceschini, CoCr alloy processed by Selective Laser Melting (SLM): Effect of Laser Energy Density on microstructure, surface morphology, and hardness, *Journal of Manufacturing Processes*, 2020, 52, p 106-119

33. T. DebRoy, H.L. Wei, J.S. Zuback, T. Mukherjee, J.W. Elmer, J.O. Milewski, A.M. Beese, A.d. Wilson-Heid, A. De, and W. Zhang, Additive manufacturing of metallic components—process, structure and properties, *Progress in materials science*, 2018, 92, p 112-224
34. F. Caiazzo, V. Alfieri, and G. Casalino, On the relevance of volumetric energy density in the investigation of Inconel 718 laser powder bed fusion, *Materials*, 2020, 13(3), p 538
35. E. ISO, Metallic materials, Vickers hardness test. Part 1: test method, 2005, p Page
36. C. Kuzu, A. Germak, C. Origlia, and E. Pelit, Result analysis of EURAMET Vickers comparison between INRiM and UME (EURAMET. MH-K1. bc), *Measurement: Sensors*, 2021, 18, p 100268
37. M. Skrzyniarz, L. Nowakowski, and S. Blasiak, Geometry, structure and surface quality of a maraging steel milling cutter printed by direct metal laser melting, *Materials*, 2022, 15(3), p 773
38. M.S. Javadi, M.V. Ehteshamfar, and H. Adibi, A comprehensive analysis and prediction of the effect of groove shape and volume fraction of multi-walled carbon nanotubes on the polymer 3D-printed parts in the friction stir welding process, *Polymer Testing*, 2023, 117, p 107844
39. M.V. Ehteshamfar, M.S. Javadi, and H. Adibi, Surface modification of prototypes in fused deposition modelling using lapping process, *Rapid Prototyping Journal*, 2022, 28(7), p 1382-1393
40. M.F. Naqibi, M. Elyasi, H.J. Aval, and M.J. Mirnia, Statistical modeling and optimization of two-layer aluminum–copper pipe fabrication by friction stir welding, *Transactions of the Indian Institute of Metals*, 2022, p 1-17
41. F. Aghabeyki, M.J. Mirnia, and M. Elyasi, Cold and warm flaring of thin-walled titanium tube using single-point incremental forming, *The International Journal of Advanced Manufacturing Technology*, 2021, 114(11), p 3357-3376
42. W.-H. Chen, M.C. Uribe, E.E. Kwon, K.-Y.A. Lin, Y.-K. Park, L. Ding, and L.H. Saw, A comprehensive review of thermoelectric generation optimization by statistical approach: Taguchi method, analysis of variance (ANOVA), and response surface methodology (RSM), *Renewable and Sustainable Energy Reviews*, 2022, 169, p 112917
43. W. Wu, J. Xue, W. Xu, H. Lin, H. Tang, and P. Yao, Parameters optimization of auxiliary gas process for double-wire SS316L stainless steel arc additive manufacturing, *Metals*, 2021, 11(2), p 190
44. S. Oberloier, W.J. Holmes, L.A. Reich, and J.M. Pearce, Particle swarm optimization of printing parameters for open-source TIG-based metal 3D printing, *Chinese Journal of Mechanical Engineering: Additive Manufacturing Frontiers*, 2022, 1(4), p 100050
45. R.R. Karri and J. Sahu, Modeling and optimization by particle swarm embedded neural network for adsorption of zinc (II) by palm kernel shell based activated carbon from aqueous environment, *Journal of environmental management*, 2018, 206, p 178-191
46. J. Kennedy and R. Eberhart, Particle swarm optimization, Proceedings of ICNN'95-international conference on neural networks, iee, 1995, p 1942-1948
47. L.C. Cagnina, S.C. Esquivel, and C.A.C. Coello, Solving engineering optimization problems with the simple constrained particle swarm optimizer, *Informatica*, 2008, 32(3),
48. M.T. Bhoskar, M.O.K. Kulkarni, M.N.K. Kulkarni, M.S.L. Patekar, G. Kakandikar, and V. Nandedkar, Genetic algorithm and its applications to mechanical engineering: A review, *Materials Today: Proceedings*, 2015, 2(4-5), p 2624-2630

49. P. Potočník, A. Jeromen, and E. Govekar, Genetic Algorithm-Based Framework for Optimization of Laser Beam Path in Additive Manufacturing, *Metals*, 2024, 14(4), p 410
50. M. Gen and R. Cheng, Genetic algorithms and engineering optimization, John Wiley & Sons, 1999
51. G.C. Onwubolu, B. Babu, and K. Deb, Introduction to genetic algorithms for engineering optimization, *New optimization techniques in engineering*, 2004, p 13-51
52. R. Hassan, B. Cohanım, O. De Weck, and G. Venter, A comparison of particle swarm optimization and the genetic algorithm, 46th AIAA/ASME/ASCE/AHS/ASC structures, structural dynamics and materials conference, 2005, p 1897
53. C. Wang, D. Li, P. Kaewniam, J. Wang, and T. Al Hababi, An ED-PSO model updating algorithm for structure health monitoring of beam-like structures, *Journal of Measurements in Engineering*, 2023, 11(3), p 358-372
54. M. Jafari, M. Hosseinzadeh, and M. Elyasi, Optimization of tube drawing process through FE analysis, intelligent computation, and experimental verification, *Proceedings of the Institution of Mechanical Engineers, Part E: Journal of Process Mechanical Engineering*, 2018, 232(1), p 94-107
55. G. Papazoglou and P. Biskas, Review and comparison of genetic algorithm and particle swarm optimization in the optimal power flow problem, *Energies*, 2023, 16(3), p 1152
56. K. Stergiou, C. Ntakolia, P. Varytis, E. Koumoulos, P. Karlsson, and S. Moustakidis, Enhancing property prediction and process optimization in building materials through machine learning: A review, *Computational Materials Science*, 2023, 220, p 112031
57. M. Carrasco, J. López, and S. Maldonado, Epsilon-nonparallel support vector regression, *Applied Intelligence*, 2019, 49(12), p 4223-4236
58. B. Ghorbani, A. Arulrajah, G. Narsilio, and S. Horpibulsuk, Experimental investigation and modelling the deformation properties of demolition wastes subjected to freeze–thaw cycles using ANN and SVR, *Construction and Building Materials*, 2020, 258, p 119688
59. E. Schulz, M. Speekenbrink, and A. Krause, A tutorial on Gaussian process regression: Modelling, exploring, and exploiting functions, *Journal of mathematical psychology*, 2018, 85, p 1-16
60. M. Farid, Data-driven method for real-time prediction and uncertainty quantification of fatigue failure under stochastic loading using artificial neural networks and Gaussian process regression, *International Journal of Fatigue*, 2022, 155, p 106415
61. J. Jenis, J. Ondriga, S. Hreck, F. Brumerick, M. Cuchor, and E. Sadovsky, Engineering applications of artificial intelligence in mechanical design and optimization, *Machines*, 2023, 11(6), p 577
62. J. Zhang, P. Wang, and R.X. Gao, Deep learning-based tensile strength prediction in fused deposition modeling, *Computers in industry*, 2019, 107, p 11-21
63. M.F. Naqibi, M. Elyasi, H.J. Aval, and M.J. Mirnia, Statistical Modeling and Optimization of Two-Layer Aluminum–Copper Pipe Fabrication by Friction Stir Welding, *Transactions of the Indian Institute of Metals*, 2022, 75(3), p 635-651
64. P.K. Ambadekar, S. Ambadekar, C. Choudhari, S.A. Patil, and S. Gawande, Artificial intelligence and its relevance in mechanical engineering from Industry 4.0 perspective, *Australian Journal of Mechanical Engineering*, 2025, 23(1), p 110-130
65. E. Yaghoubi, E. Yaghoubi, A. Khamees, and A.H. Vakili, A systematic review and meta-analysis of artificial neural network, machine learning, deep learning, and ensemble

- learning approaches in field of geotechnical engineering, *Neural Computing and Applications*, 2024, 36(21), p 12655-12699
66. A. Kumar and A.K. Ghosh, Decision tree–and random forest–based novel unsteady aerodynamics modeling using flight data, *Journal of Aircraft*, 2019, 56(1), p 403-409
 67. Y. Xu, X. Zhao, Y. Chen, and Z. Yang, Research on a mixed gas classification algorithm based on extreme random tree, *Applied Sciences*, 2019, 9(9), p 1728
 68. T. Xu, Recent advances in Rapidly-exploring random tree: A review, *Heliyon*, 2024,
 69. V. Plevris, G. Solorzano, N.P. Bakas, and M.E.A. Ben Seghier, Investigation of performance metrics in regression analysis and machine learning-based prediction models, 8th European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS Congress 2022), European Community on Computational Methods in Applied Sciences, 2022
 70. M. Naser and A.H. Alavi, Error metrics and performance fitness indicators for artificial intelligence and machine learning in engineering and sciences, *Architecture, Structures and Construction*, 2023, 3(4), p 499-517
 71. A. Tatachar, L.C. Cole, H.L. Nguyen, and K. Heinrich, Evaluation of pharmacy-based telephone interventions on medication pick-up rates: a retrospective, quality improvement study at charity outpatient clinics, *International Journal of Pharmacy Practice*, 2019, 27(6), p 510-519
 72. T.O. Hodson, Root mean square error (RMSE) or mean absolute error (MAE): When to use them or not, *Geoscientific Model Development Discussions*, 2022, 2022, p 1-10
 73. D. Chicco, M.J. Warrens, and G. Jurman, The coefficient of determination R-squared is more informative than SMAPE, MAE, MAPE, MSE and RMSE in regression analysis evaluation, *Peerj computer science*, 2021, 7, p e623
 74. T.O. Kvålseth, Cautionary note about R², *The American Statistician*, 1985, 39(4), p 279-285
 75. N.J. Nagelkerke, A note on a general definition of the coefficient of determination, *biometrika*, 1991, 78(3), p 691-692
 76. K.E. Taylor, Summarizing multiple aspects of model performance in a single diagram, *Journal of geophysical research: atmospheres*, 2001, 106(D7), p 7183-7192
 77. W. Huang, J. Yang, H. Yang, G. Jing, Z. Wang, and X. Zeng, Heat treatment of Inconel 718 produced by selective laser melting: Microstructure and mechanical properties, *Materials Science and Engineering: A*, 2019, 750, p 98-107
 78. A. Diniță, A. Neacșa, A.I. Portoacă, M. Tănase, C.N. Ilinca, and I.N. Ramadan, Additive manufacturing post-processing treatments, a review with emphasis on mechanical characteristics, *Materials*, 2023, 16(13), p 4610
 79. M.A. Buhairi, F.M. Foudzi, F.I. Jamhari, A.B. Sulong, N.A.M. Radzuan, N. Muhamad, I.F. Mohamed, A.H. Azman, W.S.W. Harun, and M. Al-Furjan, Review on volumetric energy density: influence on morphology and mechanical properties of Ti6Al4V manufactured via laser powder bed fusion, *Progress in Additive Manufacturing*, 2023, 8(2), p 265-283
 80. S.F. Nabavi, H. Dalir, and A. Farshidianfar, A comprehensive review of recent advances in laser powder bed fusion characteristics modeling: metallurgical and defects, *The International Journal of Advanced Manufacturing Technology*, 2024, 132(5), p 2233-2269

81. J. Liu, H. Ma, L. Meng, H. Yang, C. Yang, S. Ruan, D. Ouyang, S. Mei, L. Deng, and J. Chen, Laser powder bed fusion of 316L stainless steel: effect of laser polishing on the surface morphology and corrosion behavior, *Micromachines*, 2023, 14(4), p 850
82. J. Heaton, An empirical analysis of feature engineering for predictive modeling, SoutheastCon 2016, IEEE, 2016, p 1-6
83. A. Ustundag, M.S. Sivri, and K. Menguc, Feature Engineering, Business Analytics for Professionals, Springer, 2022, p Page

Figures with captions

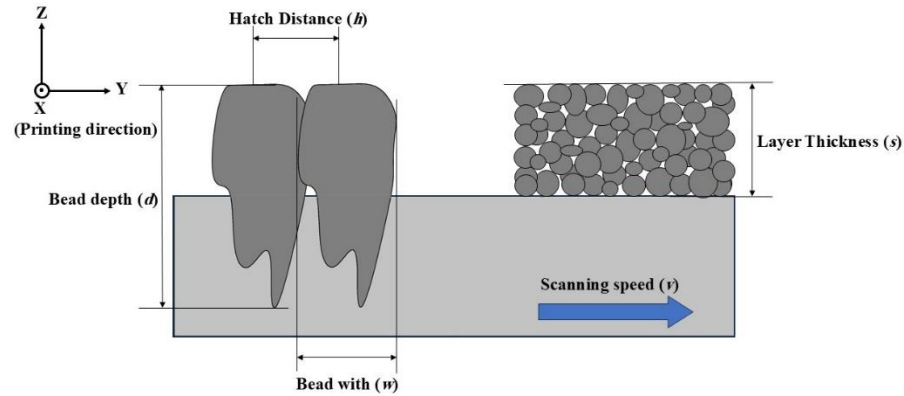


Fig. 1 The schematic representation of ED parameters

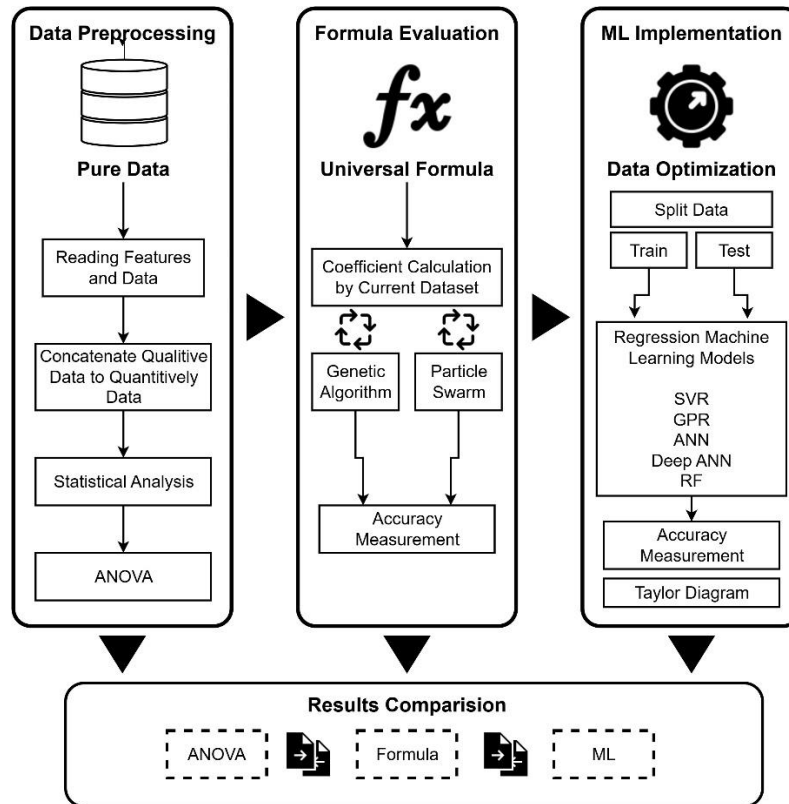
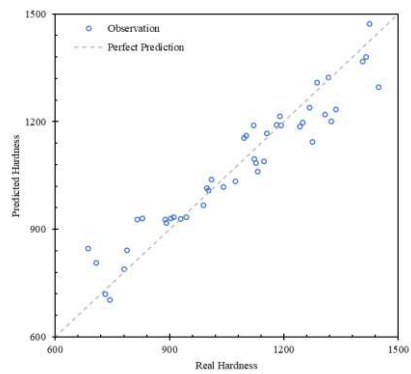
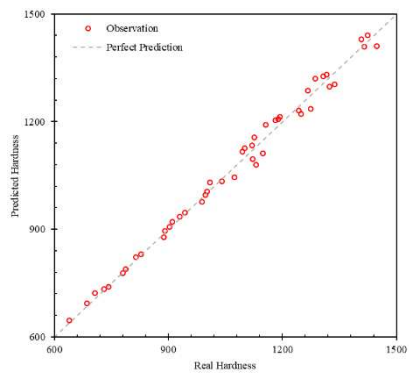


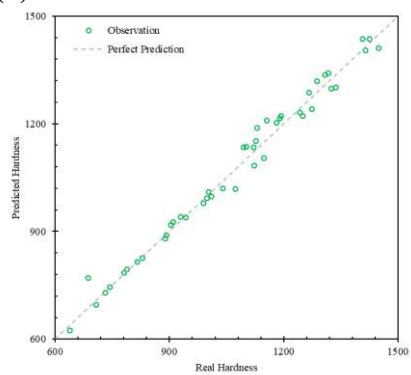
Fig. 2 The study procedure schema



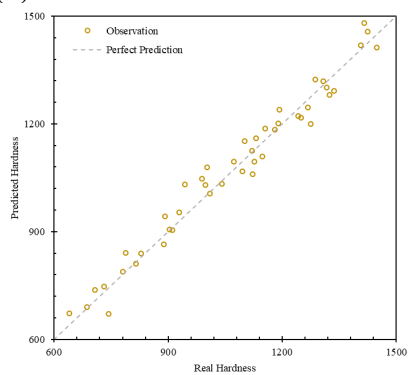
(a)



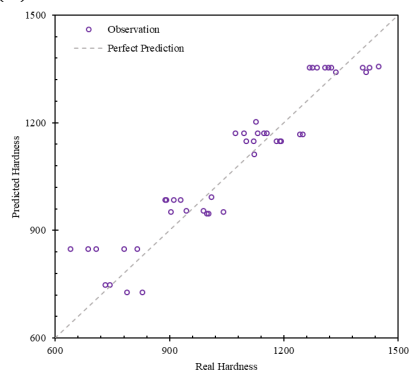
(b)



(c)

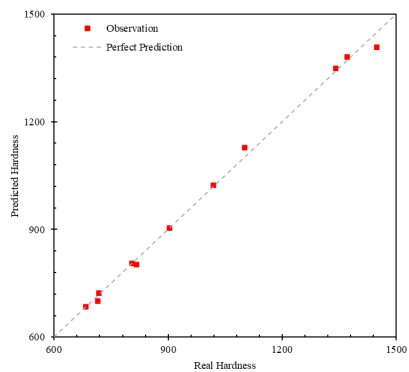
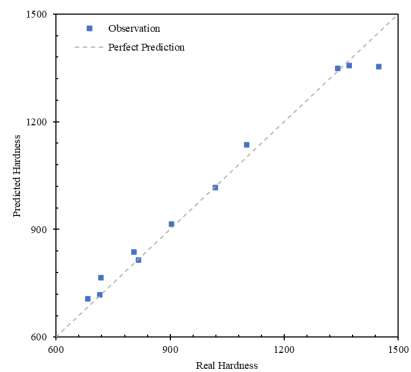


(d)



(e)

Fig. 3 Optimized training results (a) SVR, (b) GPR, (c) ANN, (d) Deep ANN, (e) RT



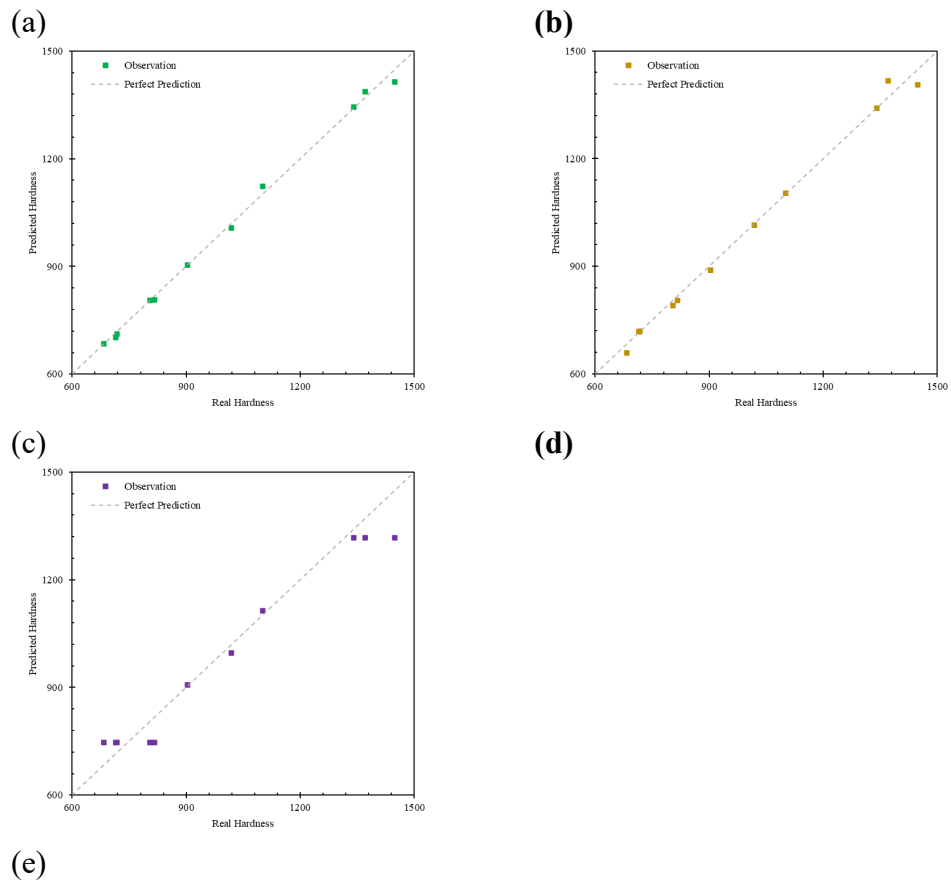
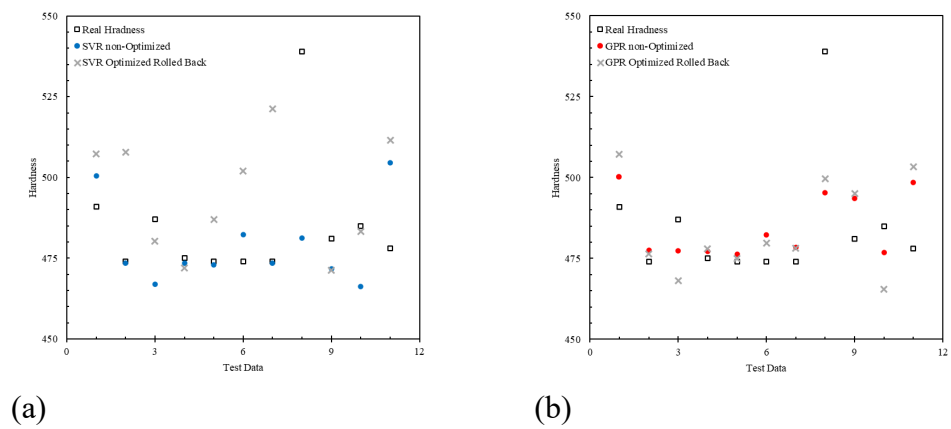


Fig. 4 Optimized testing results (a) SVR, (b) GPR, (c) ANN, (d) Deep ANN, (e) RT



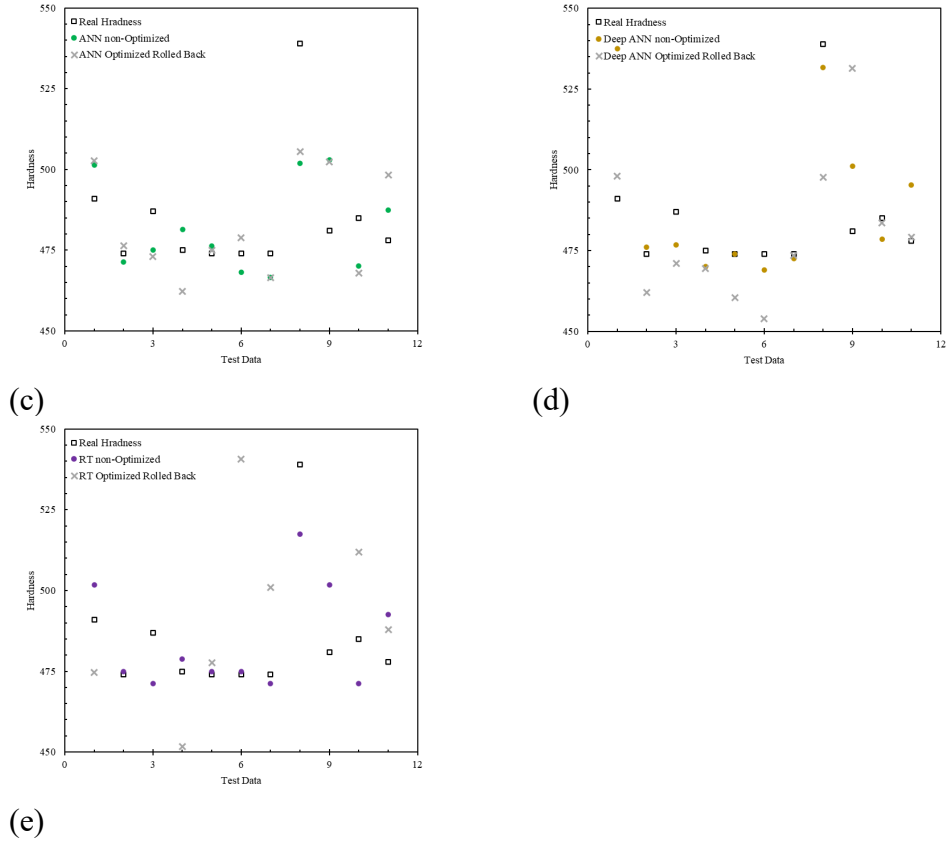


Fig. 5 Comparing experimental hardness with non-optimized and optimized roll-back by (a) SVR, (b) GPR, (c) ANN, (d) Deep ANN, (e) RT methods

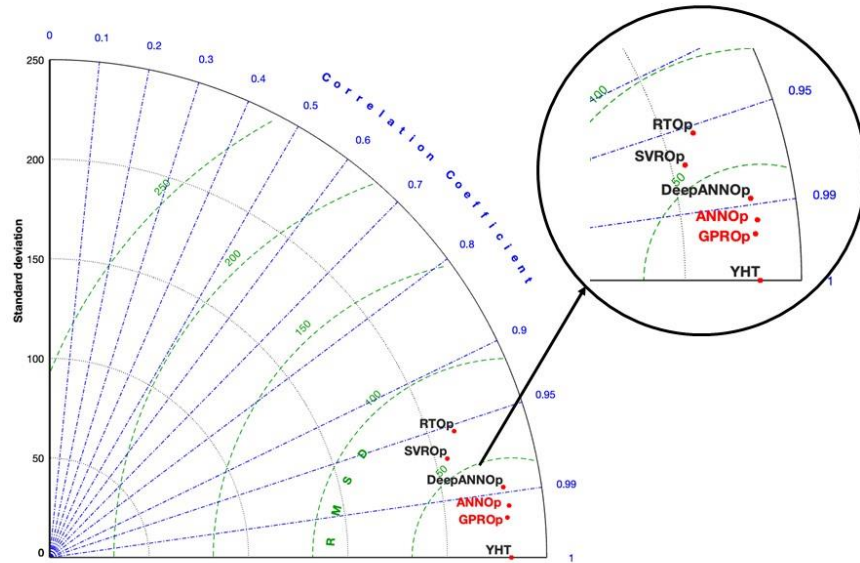


Fig. 6 Taylor diagram for all optimized ML method.

Tables with captions

Table 1: Main body and contour process parameters (Ref 14)

Title 1	Parameters	Values
Body	Layer thickness (mm)	0.03
	Spot Size (mm)	0.15
	Hatch distance (mm)	0.105
	Overlap Factor (mm)	0.7
Contour	Power (W)	192
	Spot size (mm)	0.15
	Beam compensation (mm)	0.075
	Contour Speed (mm/s)	1600
	Hatch Zone Border (mm)	0.065

Table 2: Body process parameters for different specimens (Ref 14)

Specimen Number	Laser Power (W)	Scanning Speed (mm/s)	Energy Density (J/mm ³)
1	192	400	152.38
2	192	600	101.59
3	192	800	76.19
4	230	500	146.03
5	230	600	121.69
6	230	700	104.31
7	230	900	81.13

Table 3: Process parameters and measured hardness data of selected specimens (Ref 14)

Input Parameters					Hardness	
Laser Power	Scan Speed	ED	Plane	Sample No.	As Built	Heat Treatment
192	400	152.38	ZX/ZY	D	302	491
230	600	121.69	XY	B	291	474
192	800	76.19	XY	B	284	487
230	700	104.31	XY	D	292	475
230	600	121.69	XY	C	286	474
230	500	146.03	XY	A	284	474
230	700	104.31	XY	A	282	474
230	600	121.69	ZX/ZY	D	383	539
230	500	146.03	ZX/ZY	D	327	481
192	800	76.19	XY	A	287	485
230	700	104.31	ZX/ZY	A	316	478

Table 4: Statistical Analysis of Current 56 Data in-out Dataset

	Min	Max	Average	STD
Laser Power	192	230	213.71	18.98

Inputs	Scan Speed	400	900	642.85	160.52
	Energy Density	76.19	152.38	111.90	27.73
	Plane			Qualitative Data	
Output	Sample No.			Qualitative Data	
	Heat Treated	466	541	488.63	22.38

Table 5: ANOVA Analysis Results

Parameter	Sum Sq.	d.f.	Mean Sq.	F	p-Value
Laser Power	443.6	1	443.62	1.38	0.2456
Scan Speed	4066.4	5	413.28	1.29	0.2852
Energy Density	0	0	0	0	NaN
Plane	9490	1	9490.02	29.61	0
Sample No	1113.5	3	317.16	1.16	0.3362
Error	14421.6	45	320.48		
Total	27535.1	55			

Table 6: The hyperparameters of the current particle swarm implementation

* Parameters that are automatically configured within the particle swarm algorithm.

Hyper Parameter	Value / Range	Description
Swarm Size	100	The number of particles in the swarm.
Number of Iterations	100	The maximum number of iterations or generations the algorithm will run
Inertia Weight (w)	[0.1, 1.1]	Controls the influence of the previous velocity on the current velocity
Cognitive Coefficient (c1)	1.5	The weight of the particle's own best position (personal best)
Social Coefficient (c2)	1.5	The weight of the global best position found by the swarm
Position Bounds	[0, 10]	The upper and lower bounds for the particle positions
Convergence Criteria*	1e-6 / 1000	Conditions under which the algorithm will stop, such as a maximum number of iterations or a satisfactory error threshold
Random Numbers (r1 and r2) ¹	[0, 1]	Random numbers between 0 and 1 used in the velocity update equation

Table 7: The hyperparameters of the current GA implementation

Hyper Parameter	Value / Range	Description
Population Size	100	Determines the number of individuals in each generation. A larger population size can provide a more diverse set of solutions but may increase computational cost

Number of Generations	1000	Specifies how many iterations the algorithm will run. More generations can lead to better solutions but also require more computation time
Crossover Probability	0.8	The probability that two individuals will exchange parts of their chromosomes. This helps in combining good traits from different individuals
Mutation Function	@mutationpower	Specifies how mutations are applied to individuals
Selection Function	@selectiontournament	Determines how individuals are selected for reproduction
Elite Count	10	The number of top individuals that are guaranteed to survive to the next generation. This helps in preserving the best solutions found so far

Table 8: Accuracy metrics for evaluated form formula implemented by PSO and GA

Accuracy Metrics	Particle Swarm Optimization	Genetic Algorithm
Accuracy (Max)	100%	100%
Accuracy (Min)	92%	82%
Accuracy (Mean)	97%	93%
MSE	349.2	1806.2
RMSE	18.7	42.5
MAE	15.3	35.4
MAPE	3.1%	7.2%
R ²	0.29	-2.7

Table 9: Statistical property of train and test dataset

Variable name		Min		Max		Average	
		Train	Test	Train	Test	Train	Test
Input	Laser Power	192	192	230	230	213.1	291.6
	Scan Speed	400	400	900	800	653.3	627.3
	Energy Density	76.19	76.19	152.38	152.38	109.9	115.9
	Plane	1	1	2	2	1.6	1.4
	Sample No.	1	1	4	4	2.5	2.5
Output	Heat Treated	466	474	541	539	489.7	484.7

Table 10: The all-hyper-parameters used in the implementation

ML Models	Hyperparameters	Search Space	Selected Value
SVR	C	(0.1, 1000)	13.89
	Epsilon	(0.01, 10)	1.39

	Kernel function	polynomial, gaussian, linear	Polynomial degree 3
GPR	Sigma	(0.01, 10)	4.5
	Kernel Parameter	(0.01, 10)	6.5
	Kernel function	RQ, SE, M52, Exp.	RQ
ANN	Neurons in each layer	(5, 20)	5
	Learning rate	(0.01, 1)	0.1
	Activation Function	Relu, Sigmoid, Tanh	Relu
Deep ANN	Hidden layers	(2, 10)	3
	Neurons in each layer	(5, 20)	[8, 8, 8]
	Learning rate	(0.01, 1)	0.1
	Activation Function	Relu, Sigmoid, Tanh	Relu
RT	Minimum Leaf Size	(1, 20)	5
	Maximum Number of Splits	(10, 100)	50
	Split Criterion	MSE	MSE
	Surrogate Decision	(on, off)	on
	Splits		

Table 11: Accuracy metrics for training process non-optimized parameters

Data set	Accuracy Metrics	SVR	GPR	ANN	Deep ANN	RT
Train	MSE	886.0	483.5	461.2	1658.1	522.1
	RMSE	29.8	22.0	21.5	40.7	22.8
	MAE	21.7	17.5	16.8	29.5	16.7
	MAPE	4.3	3.5	3.4	5.9	3.3
	R ²	-0.68	0.08	0.12	-2.20	0.01

Table 12: Accuracy metrics for training process optimized parameters

Data set	Accuracy Metrics	SVR	GPR	ANN	Deep ANN	RT
Train	MSE	3910.4	416.9	794.1	1505.2	5663.2
	RMSE	62.5	20.4	28.2	38.8	75.3
	MAE	47.8	16.4	21.9	31.5	62.4
	MAPE	4.7	1.5	2.1	3.0	6.5
	R ²	0.91	0.99	0.98	0.96	0.88

Table 13: Accuracy metrics for testing process optimized parameters

Data set	Accuracy Metrics	SVR	GPR	ANN	Deep ANN	RT
Test	MSE	1330.2	313.5	260.5	483.6	3211.2
	RMSE	36.5	17.7	16.1	22.0	56.7
	MAE	26.1	13.6	13.3	15.4	44.2
	MAPE	2.6	1.3	1.3	1.4	4.6
	R ²	0.98	0.99	0.99	0.99	0.95

Table 14: Results of hardness with non-optimized and optimized roll-back for different methods

HT	SVR	SVR Roll Back	GPR	GPR Roll Back	ANN	ANN Roll Back	Deep ANN	Deep Roll Back	RT	RT Roll Back
491	500.4	507.2	500.1	507.2	501.3	502.7	537.4	498.1	501.7	474.6
474	473.4	507.8	477.5	476.4	471.3	476.4	476.0	462.0	475.0	417.3
487	466.9	480.2	477.3	468.2	475.1	473.1	476.7	471.1	471.2	411.8
475	473.5	471.9	477.3	478.0	481.4	462.2	470.0	469.4	478.8	451.6
474	473.0	487.1	476.3	475.0	476.3	475.0	473.9	460.4	475.0	477.6
474	482.3	502.0	482.3	479.8	468.2	478.9	469.0	453.9	475.0	540.7
474	473.5	521.3	478.4	478.2	466.6	466.4	472.6	473.6	471.2	501.0
539	481.3	445.4	495.4	499.7	502.0	505.5	531.7	497.6	517.4	407.9
481	471.6	471.3	493.6	495.1	503.0	502.4	501.1	531.4	501.7	431.3
485	466.2	483.3	476.9	465.5	470.2	467.9	478.6	483.6	471.2	511.8
478	504.5	511.5	498.5	503.3	487.3	498.3	495.4	479.1	492.5	487.8
Mean Accuracy	97%	95%	98%	97%	98%	97%	98%	97%	98%	91%
MAE	14.0	26.1	11.3	13.6	12.2	13.3	11.0	15.4	9.7	44.2
MAPE	2.8	5.3	2.3	2.8	2.5	2.7	2.3	3.1	2.0	8.9